OPTIMAL TEST FUNCTIONS FOR K-UNSTABLE TORIC VARIETIES AND SECONDARY POLYTOPES

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Abstract. For K-unstable toric varieties, Székelyhidi's optimal test-function Θ_P is a mysterious concave function over the moment polytope P, which encodes the limit behavior of Calabi flow. This paper aims to quantize Θ_P by the maximal destabilizers for Chow-stability. We determine these destabilizers by a combinatorial object called shortest GKZ vector, which is the least norm point on the secondary polytope associated to the lattice point set $P \cap k^{-1}\mathbb{Z}^n$. When the moment polytope is Delzant, we show that these maximal Chow-destabilizers converge to Θ_P in L^2 as $k \to \infty$, without any compactness assumption. Under a C^0-precompactness assumption the convergence is uniform. It may provide an algorithm to detect K-unstability of toric varieties.

Contents

1. Introduction 2
2. Non-Archimedean quantization on toric varieties 6
2.1. Toric varieties 6
2.2. Torus-invariant NA norms 8
2.3. Berkovich analytification 8
2.4. Toric metrics 9
2.5. The super-norm map 11
2.6. Asymptotics of the volumes of super-norms 12
3. Maximum Chow-destabilizers and secondary polytopes 12
3.1. Toric reduction of the maximal Chow-destabilizers 12
3.2. Marked polytopes 13
3.3. Secondary polytopes 14
3.4. Determine Chow-stability by secondary polytopes 16
3.5. Minimize the normalized Chow-weights 17
3.6. Properties of shortest GKZ vectors 19
3.7. An example of shortest GKZ vector 20
4. Optimal test functions for K-stability 20
4.1. Székelyhidi's optimal test function Θ 20
4.2. More properties of Θ_P 22
5. From shortest GKZ vectors to optimal test-function 25
5.1. Quantum functional Ψ_k 25
5.2. Convergence problem 26
Remaining questions 35
Appendix A. Two-terms expansion for Riemann sum of convex functions 35
References 41

1. Introduction

Consider a seemingly innocent variational problem. Let P be a full-dimensional polytope in $\mathbb{R}^n$, one wants to minimize the functional

$$\Psi(\eta) = \frac{1}{2}\int_P \eta^2 \mathrm{d}x + \int_{\partial P} \eta\, \mathrm{d}\sigma,\ \ \eta : P \to \mathbb{R} \text{ is convex}, \tag{1.1}$$

where $\mathrm{d}x$ is Lebesgue measure and boundary measure $\mathrm{d}\sigma$ is proportional to the Lebesgue measure on each facet. If one decreases the boundary integral, convexity will force the L^2-norm to increase. Indeed, Ψ admits a unique minimizer f^*. In most situations, f^* is affine. It remains unknown whether f^* is always piecewise affine and how to determine it numerically.

This variational problem originates from Székelyhidi [Szé08]. Its minimizer relates to the long-time behavior of the Calabi flow on the toric Kähler manifold associated to P. We discuss this context below.

Let (X, L) be the polarized toric Kähler manifold induced by a lattice Delzant polytope $P \subset M_{\mathbb{R}} := M \otimes \mathbb{R}$, where M is a lattice of rank n. As an instance of Yau–Tian–Donaldson conjecture, the existence of extremal metrics in class $c_1(L)$ is expected to be equivalent to a stability condition on P, see [Don02; Szé14]. With the aid of torus symmetry, the existence can be reduced to the solvability of a fourth-order PDE on P. The stability condition is called the relative K-polystability, which requires that the relative Donaldson–Futaki invariant

$$\mathbb{M}_P^{\mathrm{rel}}(\eta) := \int_{\partial P} \eta\, \mathrm{d}\sigma - \int_P \eta \ell_P\, \mathrm{d}x \geq 0$$

for all rational piecewise affine convex functions η, with equality only for affine η. See §4.1 for details.

It is widely accepted that this stability condition needs to be strengthened to make the equivalence hold. There are two kinds of strengthened conditions: uniform K-stabilities [Der16] and testing more functions [Szé15]. To date, various versions of equivalence have been established, refer to [Don08; CLS18; CC21; His20; Li22; LLS23; Jub23; NS21] and their references. Meanwhile, there are partial criteria to verify stability practically, see [ZZ08; YZ19].

Optimal test function Θ_P. The unstable case has received less attention, see [WZ11; WZ14; Sek18; YZ19; Ara+23] for examples. When P is relatively K-unstable (i.e. $\mathbb{M}_P^{\mathrm{rel}}$ may be negative), [Szé08] showed the minimization problem

$$\inf\left\{ \frac{\mathbb{M}_P^{\mathrm{rel}}(\eta)}{\|\eta\|_2} \,\middle|\, 0 \neq \eta \in \mathcal{C}_1(P) \cap L^2(P) \right\},\ \ \|\eta\|_2^2 := \int_P \eta^2\, \mathrm{d}x$$

admits a unique minimizer $\ell_P - \Theta_P$ up to scaling, see (4.1) for the enlarged space $\mathcal{C}_1(P)$ of convex functions. Proposition 40 shows that $-\Theta_P$ is actually the unique minimizer of Ψ (1.1) above. Let $\Theta_P = \ell_P$ when P is relatively K-semistable (i.e. $\mathbb{M}_P^{\mathrm{rel}} \geq 0$). We call this concave function Θ_P ($= B$ in [Szé08]) the *optimal test function* for P.

The boundedness of Θ_P is proved by Li, Lian and the first author [LLS23]. Repeat their proof, we find that it can yield stronger results.

Theorem 1 ([LLS23] see §4.2)**.** *The optimal test function Θ_P can extend to a concave function on the whole space. In particular, Θ_P is continuous up to the boundary of P, and the subdifferential set $\partial(-\Theta_P)(\mathrm{int}P)$ is bounded.*

As [Szé08] showed, Θ_P encodes important information of X. For instance, if the Calabi flow admits long-time solution, then the scalar curvature (as function on P) would converge to Θ_P. More interestingly, when Θ_P is piecewise affine, it induces a polyhedral decomposition of P, then every subpolytope is semistable in a certain sense, see Theorem 38. Hence Θ_P is analogous to the Harder–Narasimhan filtration for the unstable holomorphic vector bundles, where the monotonicity of the slopes of quotient bundles mimics the concavity of Θ_P. To the authors' knowledge, there is no explicit example of Θ_P so far, see [Sek18] for examples with zero boundary measure. By the way, the optimal test-functions for Ding-stability of toric Fano varieties have been determined in [Yao22]. They are always piecewise affine.

Maximal destabilizers for K-stability. Optimal test function Θ_P is a realization of maximal K-destabilizers for general K-unstable Kähler manifold (X, L). Recall that the constant scalar curvature Kähler metrics are critical points of K-energy. When (X, L) is K-unstable, Xia [Xia21] show that there is a unique (modulo parallel relation) finite-energy geodesic ray along which the K-energy decreases fastest. By [BBJ21; Li22], this ray can be encoded into a non-Archimedean metric on L, which we call the *maximal K-destabilizer* of (X, L). In the toric setting, with the aid of symmetry, it reduces to Θ_P, see [Yao25b, Example 44].

Maximal destabilizers for Chow-stability. The second author [Yao25b] suggests to study the maximal K-destabilizers via the Chow-stability. Since its birth in [Tia97], K-stability has been closely connected with Chow-stability. Let $\iota_k : X \hookrightarrow \mathbb{P}R_k^*$ be the Kodaira embedding, where $R_k := \mathrm{H}^0(X, kL)$. Chow-stability means the GIT stability of Chow-form attached to $\iota_k(X)$. Ross and Thomas [RT06] showed that asymptotic Chow-semistability can imply K-semistability. Hence if (X, L) is K-unstable, then (X, kL) is Chow-unstable for sufficiently large k. We can consider the *maximal Chow-destabilizer* for $\iota_k(X)$, which is defined as a filtration on R_k induced by the 1-ps of $\mathrm{SL}(R_k)$ minimizing the Chow-weights.

A natural question is whether the maximal Chow-destabilizer converges to the maximal K-destabilizer in a certain sense as $k \to \infty$. This is similar to the stable case, in which Donaldson [Don01] used balanced metrics to approximate cscK metrics. One difference is that there is no Euler–Lagrange equation for maximal destabilizers, only a variational inequality.

This paper study this problem for toric varieties. Our main results are

(1) We determine the maximal Chow-destabilizers by a combinatorial object called the shortest GKZ vector, see Theorem 2 below.

(2) For a Delzant polytope, we show that the expected convergence holds in L^2 without any compactness assumption. Under a C^0-precompactness assumption the convergence is uniform, see Theorem 5 below.

Maximal Chow-destabilizers for toric varieties. For toric varieties, an explicit formula of Chow-weights (for 1-ps in the diagonal torus) is obtained in [KSZ92], see also [GKZ94; Ono13]. We will derive this formula via the NA expression of Chow-weight obtained in [Yao25b].

Let $(X^{\mathrm{an}}, L^{\mathrm{an}})$ be the Berkovich analytification of (X, L) with the trivially-normed base field $\mathbb{C}$. Let $\mathcal{N}(R_k)$ be the space of NA norms (or filtrations) on R_k. Then the Chow-weights for subvariety $\iota_k(X)$ are given by functional

$$M_k : \mathcal{N}(R_k) \to \mathbb{R},\ M_k(\chi) = \mathbb{E}\left(\mathrm{FS}_k(\chi)\right) - E_k(\chi), \tag{1.2}$$

where $\mathrm{FS}_k(\chi)$ (2.6) is the NA Fubini–Study metric induced by χ, $\mathbb{E}$ is NA Monge–Ampère energy, and $E_k(\chi)$ is the average of jumping numbers (2.14).

If $\iota_k(X)$ is Chow-unstable (i.e. $\inf M_k < 0$), the maximal Chow-destabilizer χ_k^* is the unique (up to scaling) minimizer of the functional

$$M_k(\chi)/\left\|\chi\right\|_2,\ \chi \in \mathcal{N}(R_k)\backslash\{\chi_{tr}\},$$

see [Yao25b, Theorem 1]. Moreover, χ_k^* is invariant under all automorphisms of (X, L), that allows us to search the minimizer only among the invariant norms.

Apply these facts to toric variety (X, L) given by a lattice polytope $P \subset M_{\mathbb{R}}$. It suffices to consider the space $\mathcal{N}(R_k)^T$ of torus-invariant norm on R_k. This space can be identified with

$$\mathbb{R}^{P_k} = \{f : P_k \to \mathbb{R}\}, \ P_k = P \cap k^{-1}M.$$

On the other hand, via the tropicalization map, the space $\mathrm{CPSH}_T^{\mathrm{NA}}(L)$ of invariant continuous psh metrics on L^{an} can be identified with the space $\mathcal{C}(P)$ of convex functions on P (continuous up to ∂P). Then the NA Fubini–Study map FS_k and super-norm map SN_k (2.12) are identified with the convex envelope $\vee$ (2.10) and restriction respectively.

$$\begin{tikzcd}
\mathcal{N}(R_k)^T \arrow[r, shift left, "\mathrm{FS}_k"] \arrow[d, "\simeq", "(2.4)"'] & \mathrm{CPSH}_T^{\mathrm{NA}}(L) \arrow[l, shift left, "\mathrm{SN}_k"] \arrow[d, "\simeq"', "(2.9)"] \\
\mathbb{R}^{P_k} \arrow[r, shift left, "\text{convex envelope } \vee"] & \mathcal{C}(P) \arrow[l, shift left, "\text{restriction}"]
\end{tikzcd}$$

The Chow-weight M_k (1.2) is identified with

$$M_{P_k}(f) := \frac{1}{|P_k|} \sum_{u \in P_k} f(u) - \frac{1}{V_P} \int_P f^\vee \, dx, \ \forall f \in \mathbb{R}^{P_k},$$

which is consistent with the formula in [KSZ92; Ono13].

We analyze M_{P_k} via Gelfand–Kapranov–Zelevinsky's secondary polytope $\Sigma(P_k) \subset \mathbb{R}^{P_k}$, which encodes all subdivisions of (P, P_k) and the refinement relations between them. Despite its complicated structure, we are only concerned with the least-norm point on $\Sigma(P_k)$, which we call the *shortest GKZ vector* $\sigma_k \in \Sigma(P_k)$.

Theorem 2 (see Theorem 27, 33)**.** *For toric variety (X, L) given by lattice polytope $P \subset M_{\mathbb{R}}$, suppose that $\iota_k : X \hookrightarrow \mathbb{P}R_k^*$ is an embedding ($k \geq n-1$ is enough).*

(1) Subvariety $\iota_k(X) \subset \mathbb{P}R_k^*$ is Chow-semistable (w.r.t. the action of $\mathrm{SL}(R_k)$) if and only if the constant vector $|P_k|^{-1} \in \Sigma(P_k)$, or equivalently $\sigma_k \equiv |P_k|^{-1}$.

(2) When $\iota_k(X)$ is Chow-unstable, the minimizers (i.e. maximal Chow-destabilizers) of

$$\inf \left\{ \frac{M_k(\chi)}{\|\chi\|_2} \;\middle|\; \chi \in \mathcal{N}(R_k) \backslash \{\chi_{tr}\} \right\} \tag{1.3}$$

are of the form $a\chi_k^*$ ($a > 0$), where χ_k^* is diagonalized by the basis $\{s_{u,k}\}_{u \in P_k}$ (2.3) and satisfies

$$\chi_k^*(s_{u,k}) = |P_k|^{-1} - \sigma_k(u), \ \forall u \in P_k.$$

With respect to the diagonal torus action, the equivalence in (1) has been obtained in [Ono13]. The symmetry of maximal destabilizers enables the equivalence to hold also for $\mathrm{SL}(R_k)$.

Shortest GKZ vectors for general point sets. We have not found any existing studies on the shortest GKZ vectors, which may be of independent interest for combinatorics. Thus we investigate it for general point sets that generalize P_k.

A marked polytope (Q, A) is a polytope $Q \subset M_{\mathbb{R}}$ with marking point set A such that $Q = \mathrm{conv}(A)$. A triangulation of (Q, A) gives rise to a GKZ vector $g_T : A \to \mathbb{R}$, see (3.8). The secondary polytope $\Sigma(A) \subset \mathbb{R}^A$ is the convex hull of all GKZ vectors which never contains the origin. Its faces (vertices) are in one-to-one correspondence to the regular subdivisions (triangulations) of (Q, A).

Shortest GKZ vector σ_A is the unique point on $\Sigma(A)$ with the least norm, which may not be a vertex of $\Sigma(A)$. Note that σ_A depends only on affine relations among the points in A, not on the background metric. One may think of σ_A as representing the most uniform subdivision of (Q, A) in some sense.

Theorem 3 (Theorem 34, 35)**.** *For any marked polytope (Q, A) with shortest GKZ vector $\sigma_A \in \Sigma(A)$, we have*
(1) σ_A is convex, that means the restriction of the convex envelope $\sigma_A^\vee|_A = \sigma_A$.
(2) $\sigma_A : A \to \mathbb{R}$ is positive at each point.

The convexity reminds us the concavity of Θ, see the remark after [Szé08, Proposition 9]. By (2), we know that if σ_A is a vertex of $\Sigma(A)$, then the corresponding triangulation must use all points in A.

Asymptotics of maximum Chow-destabilizers. Back to our motivation: connecting maximal K-destabilizers with maximal Chow-destabilizers. Now, the problem is reduced to connecting Θ with the shortest GKZ vectors σ_k attached to P_k.

A natural idea is to connect the variational problem behind Θ and σ_k. In appendix A, we verify the following folklore expansion of Riemann sum

$$\sum_{u\in P_k} \eta(u) = \int_P \eta\, \mathrm{d}x \cdot k^n + \frac{1}{2}\int_{\partial P} \eta\, \mathrm{d}\sigma \cdot k^{n-1} + \mathrm{o}(k^{n-1}),\ \eta \in \mathcal{C}(P),$$

which may fail if η is merely continuous. It motivates us to introduce a quantum functional

$$\Psi_k(\eta) := \frac{1}{2k^n}\sum_{u\in P_k} \eta(u)^2 + \frac{2}{k^{n-1}}\left(\sum_{u\in P_k} \eta(u) - k^n \int_P \eta\, \mathrm{d}x\right),\ \eta \in \mathcal{C}(P).$$

Actually, Ψ_k is the toric version of general functional $\breve{\mathbb{Q}}_k$ in [Yao25b]. We have

$$\lim_{k\to\infty} \Psi_k(\eta) = \Psi(\eta),\ \forall \eta \in \mathcal{C}(P).$$

Recall that $-\Theta_P$ is the unique minimizer of Ψ.

Theorem 4 (see §5.1)**.** *Quantum functional Ψ_k admits a unique minimizer*

$$\psi_k := 2V_P k^{n+1}\sigma_k^\vee - 2k \tag{1.4}$$

over $\mathcal{C}(P)$.

Its proof relies on the convexity of σ_k. The following is another main result.

Theorem 5 (see §5.2)**.** *Assume that P is Delzant. The minimizer sequence ψ_k (1.4) converges to $-\Theta_P$ in $L^2(P)$, without any precompactness assumption. More precisely,*

$$\lim_{k\to\infty} \frac{1}{k^n}\sum_{u\in P_k} \psi_k(u)^2 = \int_P \Theta_P^2\, \mathrm{d}x. \tag{1.5}$$

Consequently,

$$V_P k^n \sigma_k^\vee = 1 - \frac{\Theta_P}{2k} + \mathrm{o}_{L^2}(k^{-1}). \tag{1.6}$$

If, in addition, $\{\psi_k\}$ is C^0-precompact, then ψ_k converges uniformly to $-\Theta_P$, and hence

$$V_P k^n \sigma_k^\vee = 1 - \frac{\Theta_P}{2k} + \mathrm{o}(k^{-1}),\ k \to \infty. \tag{1.7}$$

This means that the maximal Chow-destabilizer can quantize the maximal K-destabilizer.

The L^2 statement follows from a Γ-convergence argument together with a compactness lemma for convex lattice functions. The lemma allows positive mass to concentrate near ∂P, but shows that such concentration has a nonnegative cost in the limiting functional; minimality then rules it out. By Arzelà–Ascoli theorem, C^0-precompactness is equivalent to $\{\psi_k\}$ being uniformly bounded and equicontinuous. Establishing this stronger compactness remains open.

If (1.7) holds, it may provide a method to detect the K-unstability of P. When P is K-unstable, (1.7) suggests taking $\sigma_k^\vee$ as the test function for DF-invariant $\mathbb{M}_P$. Once k is sufficiently large, we can detect K-unstability. This may be overly optimistic, since σ_k is hard to compute. There are computer programs to obtain secondary polytopes, such as [Ram02; JJK; Whi]. We write a program [Yao25a] to compute σ_k based on [Whi], yet it can only handle up to 15 points within a tolerable time on our desktop computer. We need an algorithm to find σ_k that avoids visiting the whole secondary polytope. See §5.2 for other remaining questions.

Organizations. In §2, we review the structure of Berkovich analytification of toric varieties. Translate the Fubini–Study and super-norm operators into simple operations in convex-analysis, in order to derive the formula of Chow-weights in §3.1. In §3, we study the minimization problem for Chow-weight via the secondary polytopes, and investigate the shortest GKZ vectors in §3.6. In §4, we collect some known properties of Θ_P, and then study the asymptotics of Chow-destabilizers. In appendix A, we establish a folklore expansion of Riemann sums.

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2. Non-Archimedean quantization on toric varieties

In this section, we review the non-Archimedean pluripotential theory on toric varieties. We reduce the Fubini–Study map and super-norm map to some convex-analysis operations. The main references are [BPS14; BJ24].

2.1. **Toric varieties.**

We review the structure of toric varieties, refer to [CLS11] for details.

Let M be a lattice of rank n. It induce a group algebra

$$\mathbb{C}[M] := \left\{ \text{finite sum} \sum_{m\in M} c_m \mathrm{x}^m \mid c_m \in \mathbb{C} \right\}$$

whose product is defined by $\mathrm{x}^m \cdot \mathrm{x}^{m'} = \mathrm{x}^{m+m'}$. There is a coproduct

$$\mathbb{C}[M] \to \mathbb{C}[M] \otimes \mathbb{C}[M],\ \mathrm{x}^m \mapsto \mathrm{x}^m \otimes \mathrm{x}^m,$$

which makes $\mathbb{T} := \mathrm{Spec}\mathbb{C}[M]$ a group scheme. All $\mathbb{C}$-points of $\mathbb{T}$ constitute the complex torus group $T_\mathbb{C} := \mathrm{Hom}_\mathbb{Z}(M, \mathbb{C}^*) = N \otimes_\mathbb{Z} \mathbb{C}^*$, where N is the dual lattice of M. The characters of $T_\mathbb{C}$ are given by

$$\mathrm{x}^m : T_\mathbb{C} \to \mathbb{C}^*,\ \mathrm{x}^m(n \otimes z) = z^{\langle m,n\rangle},\ n \in N,\ z \in \mathbb{C}^*.$$

The cocharacters $\mathbb{C}^* \to T_\mathbb{C}$ are given by $z \mapsto n \otimes z$, $\forall n \in N$. Let $T_\mathbb{R} := N \otimes_\mathbb{Z} \mathbb{S}^1$ be the compact torus.

A toric variety X is a complex normal variety equipped with a faithful $\mathbb{T}$-action and X has an open dense orbit. It turns out that X can be described by a combinatorial structure called a fan. A fan $\mathcal{F}$ is a finite collection of rational polyhedral convex cones in $N_{\mathbb{R}} := N \otimes_{\mathbb{Z}} \mathbb{R}$, see [CLS11, Definition 3.1.2]. It induces a toric variety $X_{\mathcal{F}}$, which is briefly reviewed below.

First, each cone $\sigma \in \mathcal{F}$ induces a finitely-generated semigroup $M_\sigma := M \cap \sigma^\vee$, where $\sigma^\vee = \{x \in M_{\mathbb{R}} \mid \langle x, \cdot \rangle \geq 0 \text{ on } \sigma\}$ is the dual cone. Then M_σ induces a finitely generated algebra $\mathbb{C}[M_\sigma]$ in the same way as $\mathbb{C}[M]$. We obtain affine variety $X_\sigma := \mathrm{Spec}\mathbb{C}[M_\sigma]$, which carries a $\mathbb{T}$-action defined by the morphism

$$\mathbb{C}[M_\sigma] \to \mathbb{C}[M] \otimes \mathbb{C}[M_\sigma],\ \mathsf{x}^m \to \mathsf{x}^m \otimes \mathsf{x}^m,\ \forall m \in M_\sigma.$$

For two cones $\sigma \subset \sigma'$ in $\mathcal{F}$, since $M_\sigma \supset M_{\sigma'}$, there is a $\mathbb{T}$-equivariant embedding $X_\sigma \hookrightarrow X_{\sigma'}$. Then glueing $\{X_\sigma \mid \sigma \in \mathcal{F}\}$ together via these embeddings, we obtain toric variety $X_{\mathcal{F}}$, which is proper if and only if $N_{\mathbb{R}}$ is the union of all cones in $\mathcal{F}$.

Let $\mathcal{F}(i) \subset \mathcal{F}$ be the subset of i-dimensional cones. These cones correspond one-to-one to the $\mathbb{T}$-orbits in $X_{\mathcal{F}}$ of dimension $n-i$, see [CLS11, §3.2]. Denote by $O(\sigma)$ the orbit corresponding to $\sigma \in \mathcal{F}$. Its Zariski closure $V(\sigma)$ is also a toric variety of dimension $n-i$, and we have $V(\sigma) = \bigsqcup_{\tau \succ \sigma} O(\tau)$. For each 1-dimensional cone $\rho \in \mathcal{F}(1)$, let $n_\rho \in N$ be the primitive generator and $D_\rho := V(\rho)$ the associated prime divisor. For any $m \in M$, x^m defines a rational function on $X_{\mathcal{F}}$ and we have

$$\mathrm{div}(\mathsf{x}^m) = \sum_{\rho \in \mathcal{F}(1)} \langle m, n_\rho \rangle\, D_\rho. \tag{2.1}$$

Let P be a full-dimensional lattice polytope in $M_{\mathbb{R}}$. It induces an (inner) normal fan

$$\mathcal{F}_P := \{\sigma_F : F \text{ is a face of } P\},$$

where

$$\sigma_F := \left\{ y \in N_{\mathbb{R}} \mid \min_P \langle \cdot, y \rangle = \langle x, y \rangle,\ \forall x \in F \right\}.$$

Let $\{n_\rho\}$ be the primitive inner normal vectors of the facets, then we have

$$P = \{x \in M_{\mathbb{R}} \mid \langle x, n_\rho \rangle + a_\rho \geq 0,\ \forall \rho \in \mathcal{F}_P(1)\},$$

where $a_\rho \in \mathbb{Z}$ since the vertices of P are lattice point.

Consider the toric variety X_P induced by the fan $\mathcal{F}_P$. Define

$$D_P := \sum_{\rho \in \mathcal{F}_P(1)} a_\rho D_\rho,$$

which is an ample and base-point-free Cartier divisor on X_P, see [CLS11, Proposition 6.1.10]. Let $L_P = \mathcal{O}_{X_P}(D_P)$ be the associated line bundle. In the next, subscript "P" will be omitted since P is fixed. We will fix a rational section s_D of L such that $\mathrm{div}(s_D) = D$. The $\mathbb{T}$-action on X can be lifted to L in a unique way such that s_D is equivariant. Let $m_\sigma \in M$ be the vertex of P corresponding to a maximal cone $\sigma \in \mathcal{F}_P(n)$. By the formula (2.1),

$$s_\sigma := \mathsf{x}^{m_\sigma} s_D \in \mathrm{H}^0(X, L) \tag{2.2}$$

is a global section and nonvanishing on X_σ. We will take s_σ as a frame of L over X_σ.

For any $k \geq 1$, we regard the rescaled lattice set

$$P_k := P \cap k^{-1} M$$

as the "quantization" of P at level k. For each $u \in P_k$, $s_{u,k} := \mathsf{x}^{ku} s_D^k$ is a global section of kL. The action of $t \in T_{\mathbb{C}}$ on $s_{u,k}$ is $t \cdot s_{u,k} = \mathsf{x}^{ku}(t^{-1}) s_{u,k}$. With respect to the $T_{\mathbb{C}}$-action, we have a weight decomposition

$$R_k := \mathrm{H}^0(X, kL) = \bigoplus_{u \in P_k} \mathbb{C} \cdot s_{u,k}. \tag{2.3}$$

In particular, $N_k := \dim R_k = |P_k| = V_P k^n + \frac{1}{2} V_{\partial P} k^{n-1} + \mathrm{O}\left(k^{n-2}\right)$.

Since kL is globally generated for any $k \geq 1$, its global sections induce a morphism $X \to \mathbb{P}R_k^*$. By [CLS11, Proposition 6.1.10], it is an embedding (kL is very ample) when $k \geq n-1$.

2.2. **Torus-invariant NA norms.** Recall a NA norm $\|\|: R_k \to \mathbb{R}_{\geq 0}$ is a norm satisfying that $\|as\| = \|s\|$ for $\forall a \in \mathbb{C}^*$ and $\|s+s'\| \leq \max\left(\|s\|, \|s'\|\right)$. We prefer to use the valuative notation $\chi = -\log\|\|$. A NA norm χ is equivalent to a left-continuous non-increasing filtration $(F_\chi^\lambda)_{\lambda\in\mathbb{R}}$ on R_k, that is $F_\chi^\lambda = \{s \in R_k \mid \chi(s) \geq \lambda\}$. Let $\mathcal{N}(R_k)$ be the space of NA norms, and $\mathcal{N}(R_k)^T$ be the subspace of $T_\mathbb{C}$-invariant NA norms, i.e. those satisfy $\chi(t\cdot s) = \chi(s)$ for any $t \in T_\mathbb{C}$ and $s \in R_k$.

Proposition 6. *A NA norm χ on R_k is $T_\mathbb{C}$-invariant if and only if χ is diagonalized by the eigen-sections $\{s_{u,k} \mid u \in P_k\}$.*

Proof. For any $\chi \in \mathcal{N}(R_k)^T$, the associated filtration $F^\lambda = \{s \in R_k \mid \chi(s) \geq \lambda\}$ is $T_\mathbb{C}$-invariant for any λ. Since R_k is multiplicity-free, F^λ is spanned by some $s_{u,k}$. This implies that χ is diagonalized by $(s_{u,k})$. The reverse direction is trivial. □

By this fact, we identify $\mathcal{N}(R_k)^T$ with the space $\mathbb{R}^{P_k} := \{f : P_k \to \mathbb{R}\}$ via the map

$$f \mapsto \chi_f, \text{ where } \chi_f\left(\sum_{u\in P_k} c_u s_{u,k}\right) = \min\left\{-kf(u) \mid c_u \neq 0\right\}. \tag{2.4}$$

There is a Tits building structure on $\mathcal{N}(R_k)$ such that $\mathcal{N}(R_k)^T$ is an apartment, see [Bou19].

2.3. **Berkovich analytification.** Let (X, L) be the polarized toric variety associated to lattice polytope P. Endowing $\mathbb{C}$ with the trivial valuation. As a set, Berkovich's analytification X^{an} is

$$X^{\mathrm{an}} := \bigcup_{\sigma\in\mathcal{F}_P} X_\sigma^{\mathrm{an}},\ X_\sigma^{\mathrm{an}} := \{v : \mathbb{C}[M_\sigma] \to \mathbb{R}\cup\{+\infty\}\},$$

where v is semivaluation extending the trivial valuation on $\mathbb{C}$. For $f \in \mathbb{C}[M_\sigma]$ and $v \in X_\sigma^{\mathrm{an}}$, we write $|f|(v) := e^{-v(f)}$. When $\sigma = \{0\}$, X_σ^{an} is the NA analytic torus $\mathbb{T}^{\mathrm{an}}$. Endowed with the Berkovich topology, X^{an} is a compact Hausdorff space. There is a kernel map $X^{\mathrm{an}} \to X$ and a center map $X^{\mathrm{an}} \to X$, which are continuous and anti-continuous respectively.

The *tropicalization* of X^{an} is a topological space N_P that is homeomorphic to P and contains $N_\mathbb{R}$ as an open dense subset, see [BPS14, §4.1]. There are two continuous maps

$$\varrho : X^{\mathrm{an}} \to N_P,\ \vartheta : N_P \to X^{\mathrm{an}}$$

called the tropicalization and the Gauss section respectively. We only give their definitions on $\mathbb{T}^{\mathrm{an}}$ and $N_\mathbb{R}$. For $v \in \mathbb{T}^{\mathrm{an}}$, i.e. a semivaluation on $\mathbb{C}[M]$, $\varrho(v) \in N_\mathbb{R}$ is defined as

$$\varrho(v) : M \to \mathbb{R},\ m \mapsto -v(\mathsf{x}^m).$$

Note that $v(\mathsf{x}^m)$ must be finite since $v(\mathsf{x}^m) + v(\mathsf{x}^{-m}) = v(1) = 0$.

For the Gauss section, given a $y \in N_\mathbb{R}$, $\vartheta_y := \vartheta(y) \in \mathbb{T}^{\mathrm{an}}$ is defined by

$$\vartheta_y\left(\sum_{m\in M} c_m \mathsf{x}^m\right) = \min_{c_m\neq 0}\langle m, -y\rangle, \text{ if } \exists c_m \neq 0.$$

It is a valuation on the function field of X. For instance, $\vartheta(0) = v_{tr}$ the trivial valuation. Moreover, the center of ϑ_y is the generic point of toric subvariety $V(\sigma_{-y})$, where $\sigma_{-y} \in \mathcal{F}_P$ is the minimal cone containing $-y$.

The image of ϑ is called the skeleton of X^{an}, denoted by $\mathrm{Sk}(X^{\mathrm{an}})$. We have $\varrho \circ \vartheta = \mathrm{Id}$, so $\mathscr{R} := \vartheta \circ \varrho$ is a retraction to the skeleton. For $v \in \mathbb{T}^{\mathrm{an}}$, we have

$$(\mathscr{R}v)\left(\sum_{m\in M} c_m \mathrm{x}^m\right) = \min_{c_m\neq 0} v(\mathrm{x}^m) \leq v\left(\sum_{m\in M} c_m \mathrm{x}^m\right). \tag{2.5}$$

$$\begin{tikzcd}
\mathbb{T}^{\mathrm{an}} \arrow[r, hook] \arrow[d, "\varrho"'] & X^{\mathrm{an}} \arrow[d, "\varrho"'] \arrow[r, hook', leftarrow] & \mathrm{Sk}(X^{\mathrm{an}}) \\
N_{\mathbb{R}} \arrow[r, hook, "\text{dense}"] & N_P \arrow[ur, "\vartheta"'] &
\end{tikzcd}$$

with $X^{\mathrm{an}} \xrightarrow{\mathscr{R}} \mathrm{Sk}(X^{\mathrm{an}})$ drawn as a bent arrow above.

Remark 7. Although $\mathbb{T}^{\mathrm{an}}$ has no usual group structure, one can define a multi-valued product and an action on X^{an}, refer to [BPS14, §4.2]. The NA compact torus $\mathbb{S}^{\mathrm{an}} \subset \mathbb{T}^{\mathrm{an}}$ is the subset of semivaluations such that $v(\mathrm{x}^m) = 0$ for all $m \in M$. Then the tropicalization N_P is the quotient space of X^{an} modulo the action of $\mathbb{S}^{\mathrm{an}}$. In the Archimedean setting, N_P is the quotient space of $X(\mathbb{C})$ modulo the action of $T_{\mathbb{R}}$. So in both settings, N_P is the common stage for toric reduction.

2.4. **Toric metrics.** Let L^{an} be the analytification of L. Since the base field $\mathbb{C}$ is trivially normed, there is a canonical metric on L^{an} called the trivial metric. Regarding the trivial metric, the norm function of a section $s \in R_k$ is $|s|_{tr}(v) := e^{-v(s)}$. When v is a divisorial valuation, $v(s)$ is just the vanishing order of s along the divisor. Any metric on L^{an} is represented by a function $\varphi : X^{\mathrm{an}} \to \mathbb{R} \cup \{-\infty\}$. With respect to metric φ, the norm of $s \in R_k$ is $|s|_\varphi := |s|_{tr} e^{-k\varphi}$.

A Fubini–Study metric φ on L^{an} is defined to have the form

$$\varphi = \frac{1}{m} \max_j \left\{\log |s_j|_{tr} + \lambda_j\right\},$$

where $m \geq 1$, $\lambda_j \in \mathbb{R}$ and $\{s_j\} \subset R_m$ are finitely many sections without common zero. A psh metric φ (not $\equiv -\infty$) on L^{an} is defined to be the pointwise limit of a decreasing sequence of Fubini–Study metrics. Denote by $\mathrm{PSH}^{\mathsf{NA}}(L)$ the set of psh metrics, and $\mathrm{CPSH}^{\mathsf{NA}}(L)$ the set of continuous psh metrics. We may omit "L" when it is fixed.

Definition 8. Suppose that (X, L) is a polarized variety and kL is globally generated. The NA Fubini–Study map $\mathrm{FS}_k : \mathcal{N}(R_k) \to \mathrm{CPSH}^{\mathsf{NA}}(L)$ is defined by

$$\mathrm{FS}_k(\chi) := \frac{1}{k} \sup_{s\in R_k\setminus\{0\}} \{\log |s|_{tr} + \chi(s)\} = \frac{1}{k} \max_{1\leq j\leq N_k} \left\{\log |s_j|_{tr} + \chi(s_j)\right\}, \tag{2.6}$$

where (s_j) is any basis diagonalizing χ.

When (X, L) is the toric variety associated to lattice polytope P, kL is always globally generated. If χ is $T_{\mathbb{C}}$-invariant, then $\mathrm{FS}_k(\chi)$ is torus-invariant in the following sense.

Definition 9. Let (X, L) be the toric variety associated to a lattice polytope $P \subset M_{\mathbb{R}}$.

(1) For any $U \subset N_P$, function $\varphi : \varrho^{-1}(U) \to \mathbb{R}$ is *torus-invariant* if $\mathscr{R}^*\varphi = \varphi$, i.e. φ is constant along the fibers of tropicalization map. Let $\mathrm{PSH}^{\mathsf{NA}}_T(L)$ ($\mathrm{CPSH}^{\mathsf{NA}}_T(L)$) be the space of torus-invariant (continuous) psh metrics on L^{an}.

(2) For a torus-invariant metric φ on L^{an}, its tropical potential $F_\varphi : N_{\mathbb{R}} \to \mathbb{R}$ is defined as

$$F_\varphi(y) := -\log |s_D|_\varphi (\vartheta_y) = \vartheta_y(s_D) + \varphi(\vartheta_y) = F_P(y) + \varphi(\vartheta_y). \tag{2.7}$$

The last $=$ is due to Proposition 10 (1) below, and $F_P(y) := \max_{u\in P} \langle u, y\rangle$ is the support function of P.

For instance, as the functions on $\mathbb{T}^{\mathrm{an}}$, $v \mapsto v(\mathrm{x}^m)$ and $|s_D|_{tr}$ are torus-invariant. Thus for $\forall u \in P_k$, the norm function $|s_{u,k}|_{tr} = \left|\mathrm{x}^{ku}\right| |s_D|_{tr}^k$ is torus-invariant. For any $\chi \in \mathcal{N}(R_k)^T$, since it is diagonalized by $(s_{u,k})$, $\mathrm{FS}_k(\chi)$ is torus-invariant.

Proposition 10. *(1) For any $y \in N_{\mathbb{R}}$, we have $\vartheta_y(s_D) = F_P(y) := \max_{u \in P} \langle u, y \rangle$.*

(2) Let $\chi \in \mathcal{N}(R_k)^T$ given by $f : P_k \to \mathbb{R}$ such that $\chi(s_{u,k}) = -kf(u)$. The tropical potential for $\mathrm{FS}_k(\chi)$ is

$$F(y) = \max_{u \in P_k} \left\{ \langle u, y \rangle - f(u) \right\}, \ \forall y \in N_{\mathbb{R}}. \tag{2.8}$$

Proof. (1) Take a full-dimensional cone $\sigma \in \mathcal{F}_P$ containing $-y$. Let m_σ be the corresponding vertex of P, then we have $\langle m_\sigma, y \rangle = \max_{u \in P} \langle u, y \rangle = F_P(y)$. Note that X_σ contains the center of valuation ϑ_y, and $s_\sigma := \mathrm{x}^{m_\sigma} s_D$ (2.2) is a frame of L over X_σ. We represent $s_D = \mathrm{x}^{-m_\sigma} s_\sigma$. By the definition of $\vartheta_y(s_D)$, we have

$$\vartheta_y(s_D) = \vartheta_y(\mathrm{x}^{-m_\sigma}) = \langle m_\sigma, y \rangle = F_P(y).$$

(2) Since χ is diagonalized by $\left\{ s_{u,k} := \mathrm{x}^{ku} s_D^k \mid u \in P_k \right\}$, we have

$$\mathrm{FS}_k(\chi) = \max_{u \in P_k} \left\{ k^{-1} \log |s_{u,k}|_{tr} - f(u) \right\}.$$

By (1), we have $\log |s_{u,k}|_{tr}(\vartheta_y) = -\vartheta_y(s_{u,k}) = \langle ku, y \rangle - kF_P(y)$. Put it into the definition (2.7), then (2.8) follows. $\square$

To capture the behavior of tropical potentials at infinity, we consider their Legendre transforms. The Legendre transform of a function $F : N_{\mathbb{R}} \to \mathbb{R}$ is defined as

$$F^*(x) = \sup_{y \in N_{\mathbb{R}}} \left\{ \langle x, y \rangle - F(y) \right\} : M_{\mathbb{R}} \to \mathbb{R} \cup \{+\infty\}.$$

Conversely, the Legendre transform of $\phi : M_{\mathbb{R}} \to \mathbb{R} \cup \{+\infty\}$ ($\not\equiv +\infty$) is defined as

$$\phi^*(y) = \sup_{x \in M_{\mathbb{R}}} \left\{ \langle x, y \rangle - \phi(x) \right\} : N_{\mathbb{R}} \to \mathbb{R} \cup \{+\infty\}.$$

Let $\mathcal{C}(P)$ be the space of convex functions on P which are continuous up to the boundary. We have a bijection

$$\{\text{convex } F : N_{\mathbb{R}} \to \mathbb{R} \mid |F - F_P| \leq C\} \xrightarrow{\simeq} \mathcal{C}(P), \ F \mapsto F^*.$$

Since for any F in the LHS, F^* is a finite-valued lsc convex function on P. By [GKR68], F^* is also upper semicontinuous, thus $F^* \in \mathcal{C}(P)$.

Theorem 11 ([BPS14, Theorem 4.8.1])**.** *Let (X, L) be the toric variety given by a lattice polytope $P \subset M_{\mathbb{R}}$. Then the map*

$$\mathrm{CPSH}_T^{\mathsf{NA}}(L) \to \mathcal{C}(P), \ \varphi \mapsto (F_\varphi)^* \tag{2.9}$$

is a bijection, where F_φ (2.7) is the tropical potential of φ.

For any $\chi \in \mathcal{N}(R_k)^T$ given by $f : P_k \to \mathbb{R}$, we define $f(x) = +\infty$ for $x \notin P_k$, then (2.8) is just f^*. Hence the Legendre transform of the tropical potential of $\mathrm{FS}_k(\chi)$ is $(f^*)^*$, which is the convex envelope of f, i.e.

$$f^\vee(x) := \sup \left\{ \eta(x) \mid \eta \in \mathcal{C}(P), \ \eta \leq f \text{ on } P_k \right\}, \ \text{for } x \in P. \tag{2.10}$$

Obviously, $f^\vee$ is a piecewise affine convex function over P. For any affine function h, we have $(f + h|_{P_k})^\vee = f^\vee + h$. For any $f_1, f_2 \in \mathbb{R}^{P_k}$, we have

$$(f_1 + f_2)^\vee \geq f_1^\vee + f_2^\vee.$$

In a summary, under (2.4) and (2.9), Fubini–Study map $\mathrm{FS}_k : \mathcal{N}(R_k)^T \to \mathrm{CPSH}_T^{\mathrm{NA}}(L)$ is identified with the convex envelope map

$$\vee : \mathbb{R}^{P_k} \to \mathcal{C}(P),\ f \mapsto f^\vee. \tag{2.11}$$

2.5. **The super-norm map.** The super-norm map is introduced in [BE21; BJ18], which plays the role of the Hilbert map in the Archimedean geometric quantization picture.

Definition 12. Let (X, L) be a polarized variety. Denote by $\mathcal{L}^\infty$ the space of bounded functions on X^{an}. The super-norm map $\mathrm{SN}_k : \mathcal{L}^\infty \to \mathcal{N}(R_k)$ is defined as

$$\mathrm{SN}_k(\varphi)(s) := -\log\left(\sup_{X^{\mathrm{an}}} |s|_{tr}\, e^{-k\varphi}\right) = \inf_{v\in X^{\mathrm{an}}} \{v(s) + k\varphi(v)\}\,. \tag{2.12}$$

In the toric setting, we show the super-norms induced by torus-invariant metrics are $T_{\mathbb{C}}$-invariant.

Proposition 13. *(1) Suppose that metric φ is torus-invariant, then for any $s \in R_k$ and $v \in X^{\mathrm{an}}$ we have $|s|_\varphi(v) \le |s|_\varphi(\mathscr{R}v)$, where $\mathscr{R} : X^{\mathrm{an}} \to \mathrm{Sk}(X^{\mathrm{an}})$ is (2.5). This implies*

$$\sup_{X^{\mathrm{an}}} |s|_\varphi = \sup_{\mathrm{Sk}(X^{\mathrm{an}})} |s|_\varphi\,. \tag{2.13}$$

(2) If bounded metric φ is torus-invariant, then the induced super-norm $\mathrm{SN}_k(\varphi)$ is $T_{\mathbb{C}}$-invariant.

Proof. (1) By (2.5) and $\varphi(v) = \varphi(\mathscr{R}v)$ for any $v \in X^{\mathrm{an}}$, we have

$$|s|_\varphi(v) = |s|_{tr}(v)e^{-k\varphi(v)} \le |s|_{tr}(\mathscr{R}v)e^{-k\varphi(\mathscr{R}v)} = |s|_\varphi(\mathscr{R}v).$$

(2) By (2.13), we only need to show that $|t\cdot s|_\varphi(v) = |s|_\varphi(v)$ for any $t \in T_{\mathbb{C}}$, $s \in R_k$ and $v \in \mathrm{Sk}(X^{\mathrm{an}})$. It amounts to show $v(t \cdot s) = v(s)$. Let $s = \sum_{u\in P_k} c_u s_{u,k}$, then $t\cdot s = \sum c_u t^{-ku} s_{u,k}$, where $t^{-ku} := \mathrm{x}^{-ku}(t) \in \mathbb{C}^*$. Take a maximal affine open set X_σ containing the center of v, and represent $s_{u,k} = \mathrm{x}^{ku-km_\sigma} s_\sigma^k$ by the local frame s_σ of L, then we have

$$v(t\cdot s) = v\left(\sum_{u\in P_k} c_u t^{-ku}\mathrm{x}^{ku-km_\sigma}\right) = \min_{c_u\neq 0} v\left(\mathrm{x}^{ku-km_\sigma}\right) = v(s),$$

where the second $=$ is due to $\mathscr{R}v = v$ and (2.5). □

By the above proposition, the super-norms induced by invariant metrics are determinated by their values on eigen-sections $s_{u,k}$.

Proposition 14. *Assume that $\varphi : X^{\mathrm{an}} \to \mathbb{R}$ is torus-invariant, bounded and upper semi-continuous, then we have $\mathrm{SN}_k(\varphi)(s_{u,k}) = -kF_\varphi^*(u)$ for all $k \ge 1$ and $u \in P_k$, where F_φ^* is the Legendre transform of the tropical potential F_φ of φ.*

Proof. For $s \in R_k$, the function $v \mapsto |s|_\varphi = e^{-v(s)-k\varphi(v)}$ is lsc on X^{an}, and $T^{\mathrm{an}} \subset X^{\mathrm{an}}$ is dense, so we have $\sup_{X^{\mathrm{an}}} |s|_\varphi = \sup_{T^{\mathrm{an}}} |s|_\varphi$. By Proposition 13 (1), it suffices to take the supremum over the skeleton, so $\sup_{T^{\mathrm{an}}} |s|_\varphi = \sup_{y\in N_{\mathbb{R}}} |s|_\varphi(\vartheta_y)$. Hence

$$\begin{aligned}\mathrm{SN}_k(\varphi)(s_{u,k}) &= \inf_{y\in N_{\mathbb{R}}} \{\vartheta_y(s_{u,k}) + k\varphi(\vartheta_y)\} = \inf_{y\in N_{\mathbb{R}}} \left\{\vartheta_y\left(\mathrm{x}^{ku}\right) + \vartheta_y\left(s_D^k\right) + k\varphi(\vartheta_y)\right\}\\ &= k\inf_{y\in N_{\mathbb{R}}} \{F_\varphi(y) - \langle u, y\rangle\} = -kF_\varphi^*(u),\end{aligned}$$

where in the second row we used $\vartheta_y(s_D) = F_P(y)$ (Proposition 10) and (2.7). □

In a summary, under (2.4) and (2.9), super-norm map $\mathrm{SN}_k : \mathrm{CPSH}_T^{\mathrm{NA}}(L) \to \mathcal{N}(R_k)^T$ is identified with the restriction map

$$\mathcal{C}(P) \to \mathbb{R}^{P_k},\ \eta \mapsto \eta|_{P_k}.$$

2.6. **Asymptotics of the volumes of super-norms.** Let (X, L) be a polarized variety. A NA norm χ on R_k corresponds to a filtration $F_\chi^\lambda = \{s \in R_k \mid \chi(s) \geq \lambda\}$ with jumping numbers:

$$\lambda_i(\chi) := \max\{\lambda \mid \dim F_\chi^\lambda \geq i\},\ 1 \leq i \leq N_k := \dim R_k.$$

If $\{s_i\} \subset R_k$ is a basis diagonalizing χ, then $\lambda_1 \geq \cdots \geq \lambda_{N_k}$ is the re-arrangement of $(\chi(s_i))_i$.

The volume of χ is defined as

$$E_k(\chi) := \frac{1}{kN_k}\sum_{i=1}^{N_k}\lambda_i(\chi). \tag{2.14}$$

For $\varphi \in \mathrm{CPSH}^{\mathsf{NA}}(L)$, Boucksom–Eriksson [BE21, Theorem A] gives the leading term of volume asymptotics, that is

$$E_k\left(\mathrm{SN}_k(\varphi)\right) = \mathbb{E}(\varphi) + \mathrm{o}(1),\ \text{ as } k \to \infty, \tag{2.15}$$

where $\mathbb{E}(\varphi)$ is the NA Monge–Ampère energy. The next-order term is expected to be the NA K-energy, which is readily verified when φ is induced by an ample test-configuration, see [Yao25b, Theorem 46].

Suppose that (X, L) is the polarized toric variety given by lattice polytope P, and $\varphi \in \mathrm{CPSH}_T^{\mathsf{NA}}(L)$ with tropical potential F_φ. By Proposition 14, we have

$$E_k\left(\mathrm{SN}_k(\varphi)\right) = -\frac{1}{|P_k|}\sum_{u\in P_k} F_\varphi^*(u).$$

Since F_φ^* is a continuous convex function over P, by the asymptotics of Riemann sum (A.2), we obtain

$$E_k\left(\mathrm{SN}_k(\varphi)\right) = -\frac{1}{V_P}\int_P F_\varphi^*\,\mathrm{d}x - \frac{1}{2V_Pk}\mathbb{M}_P\left(F_\varphi^*\right),$$

where

$$\mathbb{M}_P(\eta) := \int_{\partial P}\eta\,\mathrm{d}\sigma - \int_P \eta\bar{S}\,\mathrm{d}x,\ \bar{S} := \frac{V_{\partial P,\mathrm{d}\sigma}}{V_P}$$

is the toric NA K-energy, see §4.1 for more. Compare with (2.15), we obtain

$$\mathbb{E}(\varphi) = -\frac{1}{V_P}\int_P F_\varphi^*\,\mathrm{d}x,\ \forall\varphi \in \mathrm{CPSH}_T^{\mathsf{NA}}(L). \tag{2.16}$$

3. Maximum Chow-destabilizers and secondary polytopes

In this section, we show that the maximal Chow-destabilizer of toric variety (X_P, L_P) is determined by the shortest GKZ vector, which is the least norm point on the secondary polytope associated to point set P_k.

3.1. **Toric reduction of the maximal Chow-destabilizers.** By the torus symmetry, the minimization problem for Chow-weights can be reduced into a convex optimization problem.

Given a polarized variety (X, L), suppose k is large enough such that $\iota_k : X \hookrightarrow \mathbb{P}R_k^*$ is an embedding. Then we can talk about the Chow-stability of $\iota_k(X)$ in the sense of GIT, see [Yao25b, §4.1] for a recap of Chow-stability.

Chow-stability is usually defined via the Chow-weights assigned to the 1-ps of $\mathrm{SL}(R_k)$. The 1-ps with minimum normalized Chow-weight can differ from each other by conjugations. To eliminate this ambiguity, [Yao25b] replace 1-ps by the NA norms on R_k. Then Chow-weight turns into a functional M_k on $\mathcal{N}(R_k)$ which has a NA expression (see [Yao25b, Theorem 3])

$$M_k(\chi) = \mathbb{E}\left(\mathrm{FS}_k(\chi)\right) - E_k(\chi),\ \forall\chi \in \mathcal{N}(R_k). \tag{3.1}$$

We focus on the *Chow-unstable* case, i.e. $\inf M_k < 0$. Since M_k is homogeneous, we consider the normalized Chow-weight:

$$\bar{M}_k(\chi) := \frac{M_k(\chi)}{\|\chi\|_2},\ \text{for } \chi \in \mathcal{N}(R_k)\backslash\{\chi_{tr}\},\ \text{where } \|\chi\|_2^2 := \frac{1}{N_k}\sum_{i=1}^{N_k}\lambda_i^2(\chi).$$

By [Yao25b, Theorem 1], in the unstable case, $\bar{M}_k$ admits a unique (modulo scaling) minimizer χ_k^*, which is called the maximal Chow-destabilizer for $\iota_k(X)$. Moreover, uniqueness implies that χ_k^* inherits all symmetries of (X, L), see [Yao25b, Corollary 2].

Let (X, L) be the toric variety given by lattice polytope P. When $k \geq n-1$, ι_k is an embedding. Suppose $\iota_k(X)$ is Chow-unstable, by the symmetry principle mentioned above, the maximal Chow-destabilizer is $T_{\mathbb{C}}$-invariant. Hence, to determine it, it suffices to minimize $\bar{M}_k$ over $\mathcal{N}(R_k)^T$. Under identification $\mathcal{N}(R_k)^T \cong \mathbb{R}^{P_k}$ and (2.11), and use (2.16), we see that Chow-weight (3.1) is identified with

$$M_{P_k} : \mathbb{R}^{P_k} \to \mathbb{R},\ M_{P_k}(f) := \frac{1}{N_k}\sum_{u\in P_k} f(u) - \frac{1}{V_P}\int_P f^\vee\, \mathrm{d}x. \tag{3.2}$$

The normalized Chow-weight is identified with

$$\bar{M}_{P_k}(f) := \frac{M_{P_k}(f)}{\|f\|_2},\ \|f\|_2^2 := \sum_{u\in P_k} f(u)^2. \tag{3.3}$$

The next task is to find the minimizer of $\bar{M}_{P_k}$. We generalize this problem by replacing P_k with a more general point configuration.

3.2. **Marked polytopes.**

Definition 15. (1) Let V be a n-dimensional real vector space. A *marked polytope* in V is a pair (Q, A), where $A \subset V$ is a finite subset (mark points) and its convex hull $Q = \mathrm{conv}(A)$ is full-dimensional. Our primary example is (P, P_k).

(2) For (Q, A), we define an inner product space

$$\mathbb{R}^A := \{f : A \to \mathbb{R}\},\ \langle f, g\rangle := \sum_{u\in A} f(u)g(u).$$

The convex envelope map $\vee : \mathbb{R}^A \to \mathcal{C}(Q)$ is defined as before.

Remark 16. Even if A is not P_k, $\vee$ also relates to the NA Fubini–Study map. Let M be a lattice, and (Q, A) is a marked polytope in $M_{\mathbb{R}}$. Suppose $A = \{m_1, \cdots, m_s\} \subset M$, then the Zariski closure of the image of map

$$T_{\mathbb{C}} \to \mathbb{P}^{s-1},\ t \mapsto [\mathrm{x}^{m_1}(t) : \cdots : \mathrm{x}^{m_s}(t)]$$

is a toric variety X_A (may not isomorphic to X_Q). Take the Berkovich analytification of X_A, and consider the Fubini–Study map from the space of invariant NA norms on $\mathbb{C}^a$ to the space of NA metrics on $\mathcal{O}(1)|_{X_A}$, we will arrive at the envelope map $\vee$.

There are several interactive properties between Fubini–Study map and super-norm map, see [BE21, Lemma 7.23] and [Yao25b, Proposition 35]. The following are its toric concretizations.

Proposition 17. *Given a marked polytope* (Q, A).

(1) For $\forall f \in \mathbb{R}^A$ and $\eta \in \mathcal{C}(Q)$, we have

$$f^\vee|_A \leq f,\ (\eta|_A)^\vee \geq \eta.$$

(2) Consider the image of operation $\vee$, we have

$$\mathrm{PL}(Q,A) := \left\{ f^\vee \mid f \in \mathbb{R}^A \right\} = \left\{ \eta \in \mathcal{C}(Q) \mid (\eta|_A)^\vee = \eta \right\}.$$

Consider the restrictions of convex functions, we have

$$\mathcal{C}(A) := \{ \eta|_A \mid \eta \in \mathcal{C}(Q) \} = \left\{ f \in \mathbb{R}^A \mid f^\vee|_A = f \right\}.$$

(3) We have a bijection $\vee : \mathcal{C}(A) \to \mathrm{PL}(Q,A)$, whose inverse is the restriction map.

$$\begin{array}{ccc} \mathbb{R}^A & \underset{\text{restriction}}{\overset{\vee}{\rightleftarrows}} & \mathcal{C}(Q) \\ \cup & & \cup \\ \mathcal{C}(A) & \overset{1-1}{\longleftrightarrow} & \mathrm{PL}(Q,A) \end{array}$$

Proof. (1) is obvious. (2) For the first equality, $\supset$ is trivial. For $\forall f \in \mathbb{R}^A$, let $\eta = f^\vee$. By (1) we have $\eta|_A \le f$, so $(\eta|_A)^\vee \le f^\vee = \eta$. By (1) we also have $(\eta|_A)^\vee \ge \eta$, hence $(\eta|_A)^\vee = \eta$, so the $\subset$ direction holds. The second equality can be proved similarly. (3) follows from (2). $\square$

Definition 18. For marked polytope (Q,A), we define the "Chow-weight" functional $M_A : \mathbb{R}^A \to \mathbb{R}$ by

$$M_A(f) := \frac{1}{|A|} \sum_{u \in A} f(u) - \frac{1}{|Q|_{\mathrm{d}x}} \int_Q f^\vee \, \mathrm{d}x, \tag{3.4}$$

which generalizes (3.2). It is independent of the choices of Lebesgue measure $\mathrm{d}x$. We say (Q,A) is *Chow-semistable* if $M_A(f) \ge 0$ for all $f \in \mathbb{R}^A$. Otherwise, we say (Q,A) is *Chow-unstable*.

Suppose (Q,A) is Chow-unstable, consider the minimization problem:

$$\inf \left\{ \bar{M}_A(f) := \frac{M_A(f)}{\|f\|} \;\middle|\; 0 \not\equiv f : A \to \mathbb{R} \right\}. \tag{3.5}$$

We analyze this problem via the secondary polytopes.

3.3. **Secondary polytopes.** Roughly speaking, the secondary polytope $\Sigma(A)$ associated to a marked polytope (Q,A) is the "moduli space" of polyhedral subdivisions of Q with nodes in A. The main references are [GKZ94, §7] and [DRS10, §5].

Definition 19. Given a n-dimensional marked polytope (Q,A).

(1) A *subdivision* of (Q,A) is a family $S = \{(Q_i, A_i) \mid i \in I\}$ of n-dimensional marked polytopes such that

- $A_i \subset A$ and $Q = \bigcup_{i \in I} Q_i$.
- $\forall i,j \in I$, $Q_i \cap Q_j$ is empty or a common face of Q_i and Q_j.
- $\forall i,j \in I$, we have $A_i \cap Q_i \cap Q_j = A_j \cap Q_i \cap Q_j$.

Since $Q_i = \mathrm{conv}(A_i)$, we can simply take S as the family $\{A_i \mid i \in I\}$. Subdivision $T = \{A_i \mid i \in I\}$ is called a *triangulation* of (Q,A) if all Q_i are n-simplex and A_i is the vertex set of Q_i.

(2) We say subdivision S *refines* another one S' (denoted as $S \prec S'$, by no means strictly) if each $A_i \in S$ is contained in some $A'_j \in S'$. It defines a partial order on the set of all subdivisions of (Q,A). The unique maximum is the trivial subdivision $\{A\}$, and the triangulations are the minimum. A subdivision S is called *full* if $\bigcup_{i \in I} A_i = A$, which is equivalent to that $A_i = Q_i \cap A$ for all i.

(3) Regular subdivisions. For any $f : A \to \mathbb{R}$, its convex envelope $f^\vee$ induces a subdivision $S(f) = \{A_i \mid i \in I\}$, which is defined as follows. Let $\{Q_i\}_{i\in I}$ be the maximal subpolytopes on which $f^\vee$ is affine, and we set

$$A_i = \{u \in A \cap Q_i \mid f^\vee(u) = f(u)\}\,. \tag{3.6}$$

Subdivisions of the form $S(f)$ are called *regular*. By [DRS10, Prop. 2.2.4], (Q, A) always admits a regular triangulation. See [GKZ94, page 219] for an irregular triangulation.

A natural way to compute $\int_Q f^\vee \mathrm{d}x$ is to refine $S(f)$ by a triangulation T, then integrate $f^\vee$ over each simplex of T. This procedure yields the GKZ vectors.

Definition 20. (1) Affine interpolation. Given a triangulation T of (Q, A) and $f : A \to \mathbb{R}$, we can build a function $\Lambda_T(f)$ which is affine over each simplex of T and equals to f at the simplex vertices. Obviously, Λ_T is a linear operator, and by convexity we have

$$\Lambda_T(f) \geq f^\vee,\ \forall f : A \to \mathbb{R}. \tag{3.7}$$

(2) GKZ (Gelfand–Kapranov–Zelevinsky) vectors. For a triangulation $T = \{A_i \mid i \in I\}$ of (Q, A), the attached *GKZ vector* $g_T : A \to \mathbb{R}$ is

$$g_T(u) := \frac{1}{(n+1)\,|Q|} \sum_{i:\, u\in \mathrm{vert}(Q_i)} |Q_i|\,,\ \text{for } u \in \bigcup_{i\in I} A_i, \tag{3.8}$$

that is the sum of the volumes of simplices with u as a vertex. If $u \in A$ is not used by T, we set $g_T(u) = 0$. Note that $g_T(u)$ is independent of the choices of Lebesgue measure. The normalization factor is different from [GKZ94], our choice makes $\sum_{u\in A} g_T(u) = 1$.

Proposition 21. *For any triangulation T of (Q, A) and $f : A \to \mathbb{R}$, we have*
(1) $⨍_Q \Lambda_T(f) = \langle f, g_T\rangle$.
(2) $⨍_Q f^\vee = \langle f, g_T\rangle \;\Leftrightarrow\; f^\vee = \Lambda_T(f) \;\Leftrightarrow\; T \prec S(f)$.
(3) Let T go over all triangulations of (Q, A), we have

$$⨍_Q f^\vee = \min_T \langle f, g_T\rangle\,. \tag{3.9}$$

Proof. **(1)** Note that the integral of $\Lambda_T(f)$ over a simplex Δ of T is equal to $|\Delta|$ times the average of f over the vertices of Δ. Then take the sum over simplexes, the equality follows. **(2)** The first $\Leftrightarrow$ follows from (1) and (3.7). Suppose $T \prec S(f)$, this means that for each simplex Δ of T, there is a subpolytope (Q_i, A_i) of $S(f)$ such that $\mathrm{vert}(\Delta) \subset A_i$. By definition (3.6) of $S(f)$, $f^\vee$ is affine over $Q_i \supset \Delta$ and $f^\vee = f$ on A_i, thus $f^\vee = \Lambda_T(f)$ on Δ, so on the whole Q. Conversely, if $f^\vee = \Lambda_T(f)$, then $f^\vee$ is affine over each simplex Δ of T, and $f^\vee = \Lambda_T(f) = f$ at the vertices of Δ, these imply $T \prec S(f)$. **(3)** $\leq$ follows by (1) and (3.7). On the other hand, there exists a triangulation $T \prec S(f)$, then (2) implies (3.9). $\square$

Definition 22. The secondary polytope associated to a marked polytope (Q, A) is the following convex hull in $\mathbb{R}^A$,

$$\Sigma(A) := \mathrm{conv}\left\{g_T \in \mathbb{R}^A \mid T \text{ is triangulation of } (Q, A)\right\}.$$

Immediately, (3.9) implies

$$⨍_Q f^\vee = \min_{g\in\Sigma(A)} \langle f, g\rangle\,,\ \forall f : A \to \mathbb{R}. \tag{3.10}$$

This formula is crucial to solving minimization problem (3.5). Before that, we state the structure theorem of secondary polytopes.

Theorem 23 ([GKZ94, §7 Proposition 1.11 & Theorem 2.4])**.** *Given a n-dimensional marked polytope (Q, A).*

(1) The affine subspace spanned by $\Sigma(A)$ is

$$\left\{ g \in \mathbb{R}^A \,\middle|\, \sum_{u \in A} g(u) = 1, \sum_{u \in A} g(u) \cdot \vec{u} = \fint_Q \vec{x} \, \mathrm{d}x \right\}. \tag{3.11}$$

So $\dim \Sigma(A) = |A| - n - 1$.

(2) (Subdivision-face correspondence) For a regular subdivision S, we define

$$F(S) := \operatorname{conv} \left\{ g_T \in \mathbb{R}^A \mid \textit{triangulation } T \prec S \right\}.$$

Then the map $S \mapsto F(S)$ is an order-preserving bijection from the poset of regular subdivisions of (Q, A) to the face poset of $\Sigma(A)$. The regular triangulations are in one-to-one correspondence with the vertices of $\Sigma(A)$.

(3) For any $f \in \mathbb{R}^A$, $\langle f, \cdot \rangle |_{\Sigma(A)}$ attains its minimum exactly on face $F(S(f))$. For any regular subdivision S, the outer-normal cone of $F(S)$ is

$$\left\{ f \in \mathbb{R}^A \mid \langle f, \cdot \rangle |_{\Sigma(A)} \textit{ attains minimum on } F(S) \right\} = \left\{ f \in \mathbb{R}^A \mid S(f) \succ S \right\}.$$

Example 24. Let $A = \{(0,0), (1,0), (2,0), (1,1), (0,1)\} \subset \mathbb{R}^2$, $Q = \operatorname{conv}(A)$ is a trapezoid. Its secondary polytope $\Sigma(A)$ is a pentagon.

3.4. **Determine Chow-stability by secondary polytopes.** Define subspace

$$\mathbb{R}_0^A := \left\{ f : A \to \mathbb{R} \,\middle|\, \sum_{u \in A} f(u) = 0 \right\} \subset \mathbb{R}^A$$

and orthogonal projection

$$\pi : \mathbb{R}^A \to \mathbb{R}_0^A, \ \pi(f) = f - \bar{f}, \ \bar{f} := \frac{1}{|A|} \sum_{u \in A} f(u).$$

By (3.10), we rewrite M_A (3.4) as

$$M_A(f) = -\fint_Q (\pi f)^\vee = -\min_{g \in \Sigma(A)} \langle \pi f, g \rangle = -\min_{h \in \pi(\Sigma(A))} \langle f, h \rangle. \tag{3.12}$$

The last equality used $\langle \pi f, g \rangle = \langle f, \pi g \rangle$. Now, M_A is almost the support function of $\pi(\Sigma(A))$. By the same trick, we can handle the relative Chow-weights.

Definition 25. Given a n-dimensional marked polytope (Q, A).

(1) Let $\mathcal{A}$ be the space of affine functions. Since the restriction map $\mathcal{A} \to \mathbb{R}^A$ is injective, we can identify $\mathcal{A}$ with the image. Let $R(A) := \mathcal{A}^\perp$ be the orthogonal complement in $\mathbb{R}^A$, that is

$$R(A) = \left\{ f : A \to \mathbb{R} \,\middle|\, \sum_{u \in A} f(u) = 0, \sum_{u \in A} f(u) \, \vec{u} = 0 \right\}.$$

It records the affine relations between the points in A. Let $\Pi : \mathbb{R}^A \to R(A)$ be the orthogonal projection. By (3.11), the projection image of $\Sigma(A)$ to $\mathcal{A}$ is a single point, denoted by ℓ_A, which is the unique affine function satisfying

$$\fint_Q h = \sum_{u \in A} h(u) \ell_A(u), \ \forall \text{ affine } h.$$

(2) We define the *relative* Chow-weight functional $M_A^{\mathrm{rel}} : \mathbb{R}^A \to \mathbb{R}$ is

$$M_A^{\mathrm{rel}}(f) := -\fint_Q (\Pi f)^\vee = -\fint_Q f^\vee + \langle f, \ell_A \rangle.$$

It vanishes on affine functions. We say (Q, A) is *relatively Chow-semistable* if $M_A^{\mathrm{rel}}(f) \geq 0$ for all $f \in \mathbb{R}^A$. Otherwise, we call (Q, A) is *relatively Chow-unstable*.

Remark 26. When $(Q, A) = (P, P_k)$ and f takes rational numbers, $M_A^{\mathrm{rel}}(f)$ is the genuinely relative Chow-weight for toric variety $X_P \hookrightarrow \mathbb{P}R_k^*$ relative to the automorphism group $T_{\mathbb{C}}$, in the sense of [Szé06]. Note that our "relative Chow-semistability" differs from that in [Ono13, Definition 5.3], which involves the extremal affine function.

By (3.10), we also have support function formula

$$M_A^{\mathrm{rel}}(f) = -\min_{g \in \Sigma(A)} \langle \Pi f, g \rangle = -\min_{h \in \Pi(\Sigma(A))} \langle f, h \rangle . \tag{3.13}$$

A consequence is that the (relative) Chow-semistability is determined by the location of $\Sigma(A)$.

Theorem 27. *(1) Marked polytope (Q, A) is Chow-semistable (Definition 18) if and only if the constant function $|A|^{-1} \in \Sigma(A)$, equivalently $\ell_A \equiv |A|^{-1}$. In addition, (Q, A) is relatively Chow-semistable (Definition 25) if and only if $\ell_A \in \Sigma(A)$. Refer to figure 3.1.*

(2) For toric variety (X, L) given by a lattice polytope P, the projective embedding $\iota_k(X)$ induced by $|kL|$ is Chow-semistable if and only if $|P_k|^{-1} \in \Sigma(P_k)$, where Chow-semistability is with regard to the whole linear group action.

Proof. (1) follows from (3.12) and (3.13) respectively.

(2) By [Lee+19, Corollary 2.3, 2.7] or [Yao25b, Corollary 25], Chow-semistability is equivalent to the semistability with respect to the action of maximal torus. The latter one is the same as the Chow-semistability of (P, P_k) in the sense of Definition 18. ☐

FIGURE 3.1. Chow-semistability and the location of $\Sigma(A)$.

Remark 28. If $|P_k|^{-1} \in \Sigma(P_k)$, by (3.11), we obtain a barycenter equality

$$\frac{1}{|P_k|} \sum_{u \in P_k} \vec{u} = \frac{1}{V_P} \int_P \vec{x} \, \mathrm{d}x.$$

This is a necessary condition for $X_P \hookrightarrow \mathbb{P}R_k^*$ being Chow-semistable, which was discovered by Ono [Ono11]. Hence Chow-semistability is a very stringent property for toric varieties. In contrast, relative Chow-semistability is far easier to satisfy.

Example 29. Marked polytope (Q, A) in Example 24 is relatively Chow-semistable, since

$$\ell_A = \left(\frac{5}{27}, \frac{5}{27}, \frac{5}{27}, \frac{2}{9}, \frac{2}{9} \right) \in \Sigma(A),$$

which is the restriction of affine function $h(x, y) = \frac{1}{27}(y + 5)$.

3.5. **Minimize the normalized Chow-weights.** Suppose (Q, A) is Chow-unstable, we determine the minimizers of $\bar{M}_A$ (3.5). It follows from the basic fact below.

Lemma 30. *Let K be a compact convex subset in an inner-product space W. Suppose $0 \notin K$, let $x^* \in K$ be the unique point on K with minimal norm.*

(1) $\langle x^*, x \rangle \geq \langle x^*, x^* \rangle$ for all $x \in K$. Moreover, x^* is the unique point on K satisfying this property.

(2) Let $H(x) = \min_{y \in K} \langle x, y \rangle$, then we have

$$\max_{x \in W \setminus \{0\}} \frac{H(x)}{\|x\|} = \min_{y \in K} \|y\| = \|x^*\| . \tag{3.14}$$

The maximizers for the left-hand side are $\{ax^* \mid a > 0\}$.

Proof. (1) For any $x \in K$, let $x_t := tx + (1-t)x^* \in K$ for $t \in [0,1]$. Since x^* has minimal norm, we have

$$0 \leq \frac{d}{dt} \|x_t\|^2 |_{t=0} = 2 \langle x - x^*, x^* \rangle .$$

Conversely, if $y \in K$ satisfies this condition. It implies $\|y\|^2 \leq \langle y, x^* \rangle \leq \|y\| \|x^*\|$, so $\|y\| \leq \|x^*\|$. Since the minimal-norm point is unique, we have $y = x^*$.

(2) For any $x \in W \backslash \{0\}$ and $y \in K$, Cauchy–Schwarz inequality $\langle x, y \rangle \leq \|x\| \|y\|$ implies

$$\frac{\langle x, y \rangle}{\|x\|} \leq \|y\| .$$

Take $\inf_{y \in K}$ on both sides, we obtain $\frac{H(x)}{\|x\|} \leq \min_{y \in K} \|y\| = \|x^*\|$. Hence $\leq$ side of (3.14) follows. On the other hand, by (1), we have

$$\frac{H(x^*)}{\|x^*\|} = \frac{\min_{y \in K} \langle x^*, y \rangle}{\|x^*\|} = \frac{\langle x^*, x^* \rangle}{\|x^*\|} = \|x^*\| .$$

Hence (3.14) holds and x^* is a maximizer of $\bar{H}(x) := H(x) / \|x\|$.

Uniqueness. Suppose x_0, x_1 are both maximizers for $\bar{H}$. Scaling them such that $\|x_0\| = \|x_1\| = 1$, then $H(x_0) = H(x_1) = \max \bar{H} > 0$. Consider $x_m = (x_0 + x_1)/2 \neq 0$. If $x_0 \neq x_1$, then we have $\|x_m\| < 1$. Since H is concave,

$$H(x_m) \geq \frac{1}{2} (H(x_0) + H(x_1)) = \max \bar{H}.$$

This follows that

$$\bar{H}(x_m) = \frac{H(x_m)}{\|x_m\|} \geq \frac{\max \bar{H}}{\|x_m\|} > \max \bar{H}.$$

That is absurd, so $x_0 = x_1$. □

Definition 31 (shortest GKZ vector)**.** For a marked polytope (Q, A), since $0 \notin \Sigma(A)$, we define the *shortest GKZ vector* $\sigma_A \in \Sigma(A)$ is the unique least-norm point on $\Sigma(A)$. By the subdivision-face correspondence (Theorem 23), there is a unique regular subdivision S_A of (Q, A) such that $F(S_A)$ is the minimal face containing σ_A. Refer to figure 3.2.

Remark 32. Note that σ_A may not be a vertex of $\Sigma(A)$ (i.e. S_A is not a triangulation). In that case, σ_A is a convex combination of GKZ vectors. See §3.7 for an example.

FIGURE 3.2. The secondary polytope $\Sigma(A)$ and the shortest GKZ vector σ_A.

Now, we determine the maximal Chow-destabilizers by shortest GKZ vectors.

Theorem 33. *Suppose marked polytope (Q, A) is Chow-unstable (i.e. $\inf M_A < 0$), then we have*

$$\min \left\{ \frac{M_A(f)}{\|f\|} \;\middle|\; 0 \not\equiv f : A \to \mathbb{R} \right\} = - \left\| \sigma_A - |A|^{-1} \right\| .$$

The minimizers are $a \left(\sigma_A - |A|^{-1} \right)$, $\forall a > 0$. Applying to the case $(Q, A) = (P, P_k)$, we obtain the part (2) of Theorem 2.

Proof. Apply Lemma 30 to the formula (3.12) of M_A. Take $K = \pi\left(\Sigma(A)\right)$ and the least-norm point is $\sigma_A - |A|^{-1}$.

For part (2) of Theorem 2. Suppose (X, L) is the toric variety induced by P and $\iota_k(X)$ is Chow-unstable. By the symmetry principle [Yao25b, Corollary 2], the minimizers of (1.3) belong to $\mathcal{N}(R_k)^T$ over which the Chow-weight has formula (3.3). Then the statement follows. □

3.6. **Properties of shortest GKZ vectors.** We establish two properties of σ_A, the convexity and positivity.

Theorem 34. *Let $\sigma_A : A \to \mathbb{R}$ be the shortest GKZ vector associated to marked polytope (Q, A), then $\sigma_A^\vee|_A = \sigma_A$, which means that σ_A is convex.*

Proof. Consider functional:

$$H(f) := \fint_Q f^\vee = \min_{g\in\Sigma(A)} \langle f, g\rangle\,,\ f : A \to \mathbb{R}.$$

By Lemma 30, maximizers of $\bar{H}(f) := H(f)/\|f\|$ are $a\sigma_A$, $\forall a > 0$.

Let $\rho := \sigma_A^\vee|_A$, we have $0 \le \rho \le \sigma_A$, thus $\|\rho\| \le \|\sigma_A\|$. By Proposition 17 (2), we have $\rho^\vee = \sigma_A^\vee$. These imply

$$\frac{H(\rho)}{\|\rho\|} = \frac{H(\sigma_A)}{\|\rho\|} \ge \frac{H(\sigma_A)}{\|\sigma_A\|}.$$

Hence ρ is also a maximizer of $\bar{H}$. We have $\rho = a\sigma_A$ for some $a > 0$. But $\rho^\vee = \sigma_A^\vee$ implies $a = 1$, so $\rho = \sigma_A$. □

It is obvious that $\sigma_A \ge 0$ from its definition, further we can show that $\sigma_A > 0$ at every point.

Theorem 35. *Let (Q, A) be a marked polytope.*

(1) Let S_A be the unique regular subdivision such that $F(S_A)$ is the minimal face of $\Sigma(A)$ containing σ_A, then we have $S(\sigma_A) \succ S_A$. We cannot exclude the possibility that $S(\sigma_A) \ne S_A$. Conversely, if $g \in \Sigma(A)$ satisfies $g \in F\left(S(g)\right)$, then $g = \sigma_A$.

(2) We have $\sigma_A > 0$ at every point in A. Subdivision S_A must be full, so is $S(\sigma_A)$, which implies $\sigma_A^\vee|_A = \sigma_A$.

Proof. (1) By Lemma 30 (1), function $\langle \sigma_A, \cdot\rangle\,|_{\Sigma(A)}$ attains its minimum at σ_A, so on the whole face $F(S_A)$, since σ_A is contained in the relative interior of $F(S_A)$. On the other hand, Theorem 23 (3) says that the minimizer set is face $F\left(S(\sigma_A)\right)$. Hence $F\left(S(\sigma_A)\right) \supset F(S_A)$, so $S(\sigma_A) \succ S_A$. For the last statement, $g \in F\left(S(g)\right)$ implies that $\langle g, \cdot\rangle\,|_{\Sigma(A)}$ attains its minimum at g, by Lemma 30 (1), we can conclude $g = \sigma_A$.

(2) First we handle a simple case, namely σ_A is a vertex of $\Sigma(A)$, then $\sigma_A = g_T$ for a regular triangulation T and $S_A = T$. By (1), we have $S(\sigma_A) \succ T$. Use Proposition 21 (2), we have

$$\sigma_A^\vee = \Lambda_T(\sigma_A) = \Lambda_T(g_T).$$

By definition of GKZ vector, $g_T > 0$ at every vertex of T, thus $\Lambda_T(g_T) > 0$ on Q, i.e. $\sigma_A^\vee > 0$. Since $\sigma_A \ge \sigma_A^\vee|_A$, it implies $\sigma_A > 0$ at every point.

General case. Since $\sigma_A \in F(S_A)$, there exists triangulations $T_i \prec S_A$ $(1 \le i \le p)$ such that

$$\sigma_A = \sum_{i=1}^p \lambda_i g_{T_i},\ \sum_{i=1}^p \lambda_i = 1,\ \lambda_i > 0.$$

Since $S(\sigma_A) \succ S_A \succ T_j$, by Proposition 21 (2), we have

$$\sigma_A^\vee = \Lambda_{T_j}(\sigma_A),\ \forall 1 \le j \le p.$$

Since operator Λ_{T_j} is linear, for each j we have

$$\Lambda_{T_j}(\sigma_A) = \sum_{i=1}^p \lambda_i \Lambda_{T_j}(g_{T_i}) \ge \lambda_j \Lambda_{T_j}(g_{T_j}),$$

where we used $\Lambda_{T_j}(g_{T_i}) \ge 0$. Since $\Lambda_{T_j}(g_{T_j}) > 0$ on Q, so $\sigma_A^\vee = \Lambda_{T_j}(\sigma_A) > 0$. Since $\sigma_A \ge \sigma_A^\vee|_A$, we obtain $\sigma_A > 0$ at every point.

Next we show S_A is full. For any $a \in A$, we have $0 < \sigma_A(a) = \sum_{i=1}^{p} \lambda_i g_{T_i}(a)$, so there exists j such that $g_{T_j}(a) > 0$. By definition of GKZ vector, it means that a is a vertex of T_j. Since $T_j \prec S_A$, a is also a mark point of S_A, thus S_A is full. Then $S(\sigma_A)$ is also full since $S(\sigma_A) \succ S_A$. $\square$

Remark 36. Notice that $\sigma_A > 0$ indicates to us how complicated σ_A is. Represent $\sigma_A = \sum_{i=1}^{p} \lambda_i g_{T_i}$ $(\sum \lambda_i = 1, \lambda_i > 0)$ by some triangulations T_i of (Q, A), then $\sigma_A > 0$ implies that every point in A is a vertex of some T_i. Furthermore, if $\sigma_A = g_T$ is a vertex of $\Sigma(A)$, then T must use all the points in A.

3.7. **An example of shortest GKZ vector.** Let

$$A = \{(0,0), (0,2), (2,0), (2,2), (1,1), (6,5)\} \subset \mathbb{R}^2, \ Q = \mathrm{conv}(A).$$

There is a SageMath function [Whi] can output $\Sigma(A)$. It turns out $\Sigma(A)$ is 3-dimensional and has 7 facets, 15 edges and 10 vertices, see its 3Dview. Based on [Whi], we write a code [Yao25a] to compute the shortest GKZ vector σ_A. The result is

$$\sigma_A = \left(\frac{15}{121}, \frac{5}{33}, \frac{493}{3025}, \frac{1511}{9075}, \frac{1318}{9075}, \frac{2267}{9075} \right).$$

It shows that (Q, A) is relatively Chow-unstable, and σ_A lies in the relative interior of a facet. In this example, we have $S(\sigma_A) = S_A$. Figure 3.3 shows subdivision S_A.

FIGURE 3.3. The regular subdivision S_A induced by the shortest GKZ vector.

4. OPTIMAL TEST FUNCTIONS FOR K-STABILITY

4.1. **Székelyhidi's optimal test function Θ.** We recap the optimal test-function for K-unstable toric varieties in [Szé08]. For an introduction to K-stability of toric varieties, refer to [Don02; Szé06; WZ14].

Let $P \subset M_{\mathbb{R}}$ be a lattice polytope. Given a rational piecewise affine convex function η over P, one can build a toric test-configuration for (X_P, L_P), see [Don02]. Donaldson shows that the attached DF-invariant is equal to

$$\mathbb{M}_P(\eta) := \int_{\partial P} \eta \, \mathrm{d}\sigma - \int_P \eta \bar{S} \, \mathrm{d}x, \ \bar{S} = \frac{V_{\partial P, \mathrm{d}\sigma}}{V_P},$$

see (A.2) for the definition of $\mathrm{d}\sigma$. When η is affine, it recovers the classical Futaki invariant with respect to the automorphism torus action.

Since we focus on the unstable case, we only recall the K-semistability and its relative version. The relative DF-invariant is defined as

$$\mathbb{M}_P^{\mathrm{rel}}(\eta) := \int_{\partial P} \eta \, \mathrm{d}\sigma - \int_P \eta \ell_P \, \mathrm{d}x,$$

where ℓ_P is the unique affine function such that $\mathbb{M}_P^{\mathrm{rel}}$ vanishes on all affine functions. Conventionally, ℓ_P is called the *extremal affine function*, since it can give the Hamiltonian function for the extremal vector field.

Definition 37 (K-semistabilities)**.** Given a lattice polytope $P \subset M_{\mathbb{R}}$.

(1) we say P is K-semistable if $\mathbb{M}_P(\eta) \geq 0$ for all $\eta \in \mathcal{C}(P)$. Otherwise, we say P is K-unstable.

(2) we say P is relatively K-semistable if $\mathbb{M}_P^{\mathrm{rel}}(\eta) \geq 0$ for all $\eta \in \mathcal{C}(P)$. Otherwise, we say P is relatively K-unstable.

In the K-unstable case, Székelyhidi [Szé08] considered the minimization problem for the normalized invariant

$$\bar{\mathbb{M}}_P(\eta) := \frac{\mathbb{M}_P(\eta)}{\|\eta\|_2}, \quad \|\eta\|_2^2 = \int_P \eta^2,$$

and obtained a minimizer in an enlarged space. Let P^* equals to P excluding all the faces with codimension ≥ 2, which is also a convex set. Consider space

$$\mathcal{C}_1(P) = \left\{ \eta : P^* \to \mathbb{R} \text{ continuous convex} \;\middle|\; \int_{\partial P} \eta \, \mathrm{d}\sigma < \infty \right\}. \tag{4.1}$$

Functions in $\mathcal{C}_1(P)$ can be extended to lsc convex functions on the whole P (may take $+\infty$). By [Don02, Lemma 5.1.3], they are integrable over P, thus $\mathbb{M}_P$ is well-defined on $\mathcal{C}_1(P)$.

Theorem 38 ([Szé08])**.** *Given a lattice polytope $P \subset M_{\mathbb{R}}$.*

(1) Suppose P is K-unstable, then

$$\inf \left\{ \bar{\mathbb{M}}_P(\eta) \mid 0 \neq \eta \in \mathcal{C}_1(P) \cap L^2(P) \right\} = -\left\| \Theta_P - \bar{S} \right\|_2$$

admits a unique (up to scaling) minimizer $\bar{S} - \Theta_P \in \mathcal{C}_1(P) \cap L^2(P)$. We call concave function Θ_P the optimal test-function for P (denoted by B in [Szé08]), which satisfies

$$\int_{\partial P} \eta \geq \int_P \eta \Theta_P, \ \forall \eta \in \mathcal{C}_1(P) \cap L^2(P) \text{ and } \int_{\partial P} \Theta_P = \int_P \Theta_P^2. \tag{4.2}$$

When P is K-semistable, we define $\Theta_P = \bar{S}$, above conditions still hold.

When P is relatively K-semistable, we have $\Theta_P = \ell_P$. If P is relatively K-unstable, $\ell_P - \Theta_P$ is the unique minimizer of $\mathbb{M}_P^{\mathrm{rel}}(\eta) / \|\eta\|$.

(2) Consider the space of "scalar curvatures"

$$\mathcal{S}(P) = \left\{ \theta \in L^2(P) \;\middle|\; \int_{\partial P} \eta \geq \int_P \eta\theta, \ \forall \eta \in \mathcal{C}_1(P) \cap L^2(P) \right\}, \tag{4.3}$$

then Θ_P is the unique element in $\mathcal{S}(P)$ with minimal L^2-norm. Recall that if $u : P \to \mathbb{R}$ is the symplectic potential of an invariant Kähler metric, its scalar curvature is $S(u) = -\sum_{i,j} \partial_i \partial_j u^{ij} \in \mathcal{S}(P)$.

(3) Consider the Calabi flow on X_P with invariant initial data, which can be reduced into evolution equation on P:

$$\dot{u}_t = -S(u_t) = \sum_{i,j} \partial_i \partial_j u_t^{ij}.$$

Assume it admits long-time solutions, then $S(u_t) \xrightarrow{L^2} \Theta_P$ as $t \to \infty$.

(4) (canonical semi-stable decomposition) Assume P is relatively K-unstable and Θ_P is piecewise affine. Let $Q \subset P$ be a maximal subpolytope over which Θ_P is affine, then Q is relatively K-semistable in the following sense:

$$\int_{\partial Q \cap \partial P} \eta \, \mathrm{d}\sigma \geq \int_Q \eta \Theta_P \, \mathrm{d}x, \ \forall \eta \in \mathcal{C}(Q). \tag{4.4}$$

Remark 39 (no island)*.* Under the assumptions in (4), we can deduce that each subpolytope Q must have a facet which is entirely contained in ∂P. If not, we have $|\partial Q \cap \partial P|_{\mathrm{d}\sigma} = 0$. Take η in (4.4) to be the affine functions extending $\pm\Theta|_Q$, it implies

$$\int_Q \Theta^2 = \int_{\partial Q \cap \partial P} \Theta \, \mathrm{d}\sigma = 0.$$

So $\Theta \equiv 0$ on Q. Since Θ is concave, this implies $\Theta \leq 0$ on P, but this contradicts with $\int_P \Theta = V_{\partial P, \mathrm{d}\sigma} > 0$, which follows from (4.2).

4.2. **More properties of Θ_P.** A central question is whether Θ is piecewise affine. The answer is still unknown. A closely related result is Theorem 42. In this section, we give some properties about Θ. First, we re-characterize Θ as the unique minimizer of a simpler convex functional, which is inspired by the work [Ino22] of Inoue.

Proposition 40. *Convex function $-\Theta_P$ is the unique minimizer of functional*

$$\Psi(\eta) := \frac{1}{2}\int_P \eta^2 \mathrm{d}x + \int_{\partial P} \eta \, \mathrm{d}\sigma, \ \eta \in \mathcal{C}_1(P) \cap L^2(P). \tag{4.5}$$

Proof. For any $\eta \in \mathcal{C}_1(P) \cap L^2(P)$, (4.2) implies

$$\int_{\partial P} \eta \geq \int_P \eta\Theta \geq -\|\eta\|_2 \|\Theta\|_2 \geq -\frac{1}{2}\|\eta\|_2^2 - \frac{1}{2}\|\Theta\|_2^2 .$$

Thus $\Psi(\eta) \geq -\frac{1}{2}\|\Theta\|_2^2$. This lower bound is achieved by $\eta = -\Theta$ since $\int_P \Theta^2 = \int_{\partial P}\Theta$. The uniqueness follows from the strict convexity of L^2-norm. $\square$

Proposition 41. *We have* $\sup_P \Theta_P = \sup_{\partial P} \Theta_P > 0$.

Proof. Set $u = -\Theta$, we show $\inf_P u = \inf_{\partial P} u$. If this is not true, let $m := \inf_P u < \inf_{\partial P} u$. Since $\int_P \Theta = |\partial P|_{\mathrm{d}\sigma} > 0$, we know $m < 0$. Take $\delta > 0$ such that $0 > m + \delta < \inf_{\partial P} u$. Consider $u' = \max\{u, m+\delta\}$. Since $u \geq \inf_{\partial P} u > m + \delta$ on ∂P, we have $u' = u$ on ∂P. Then it is easy to see

$$\Psi(u') = \frac{1}{2}\int_P u'^2 + \int_{\partial P} u' < \frac{1}{2}\int_P u^2 + \int_{\partial P} u = \Psi(u).$$

It is absurd since u is a minimizer of Ψ. $\square$

In [LLS23], the authors showed the boundedness of Θ_P. We find that their proof can yield stronger results. For readers' convenience, we reproduce the proof with some simplifications.

Proof of Theorem 1. For readers to make comparisons, we use the same notations as [LLS23, Theorem 6.2]. For example, coordinates on $N_{\mathbb{R}}$ are denoted by x, and ξ for $M_{\mathbb{R}}$.

Translate P such that $0 \in \mathrm{int}P$, then $a_\rho > 0$. Let

$$u := -\Theta_P : P \to \mathbb{R} \cup \{+\infty\},$$

which is lsc, convex and taking finite values on P^*. We set $u = +\infty$ outside P. Fix any $p^o \in \partial u(0)$, let $\ell(\xi) := u(0) + \langle p^o, \xi\rangle$ be the attached support function. Define $u^o := u - \ell$, we have $u^o(\xi) \geq u^o(0) = 0$ for $\forall \xi \in P$. Let

$$f^o(x) := (u^o)^*(x) := \sup_{\xi \in P} \langle x, \xi\rangle - u^o(\xi), \ x \in N_{\mathbb{R}}$$

be the Legendre dual of u^o. We also have $f^o(x) \geq f^o(0) = 0$. Since $0 \in \mathrm{int}P$, there exists a closed ball $B \subset \mathrm{int}P$ centered at 0, then we have

$$I_P \leq u^o \leq I_B + \sup_B u^o,$$

where I_B is the convex characteristic function of B, i.e. $I_B = 0$ on B and $= +\infty$ otherwise. Taking the Legendre dual, we obtain

$$\sup_{\xi \in P} \langle x, \xi\rangle \geq f^o(x) \geq \sup_{\xi \in B}\langle x, \xi\rangle - \sup_B u^o, \ x \in N_{\mathbb{R}}.$$

It implies that the sublevel set

$$S_{f^o}(h) := \{x \in N_{\mathbb{R}} \mid f^o(x) \leq h\}$$

is compact and convex.

We introduce the key construction in [LLS23, before Lem. 3.9]. For $h > 0$, we define the *Legendre truncation* of u^o is

$$u_h^o(\xi) := \sup_{x : f^o(x) \leq h} \langle x, \xi \rangle - f^o(x) = \left((u^o)^* + I_{S_{f^o}(h)}\right)^* (\xi).$$

It is the tropical analogue of the Fourier truncation. Intuitionally, for $h \gg 1$, u_h^o is the flattening of u over the region with large slope. Since $f^o \geq 0$, it is easy to see

$$\sup_{f^o(x) \leq h} \langle x, \xi \rangle - h \leq u_h^o \leq u^o, \quad \sup_{f^o(x) \leq h} \langle x, \xi \rangle . \tag{4.6}$$

Set

$$u_h := u_h^o + \ell \leq u^o + \ell = u,$$

which is nondecreasing in h. We will show that

$$u_h = u, \text{ on } P \text{ when } h \gg 1.$$

The proof proceeds in many steps. For $h > 0$, let

$$W_h := \partial f^o \{f^o \leq h\} \subset P,$$

which is compact by [Fig17, Lemma A.22], and may not be convex.

(a) We show $u_h^o = u^o$ on W_h. Suppose $\xi \in \partial f^o(x)$ with $f^o(x) \leq h$, then we have

$$u_h^o(\xi) \geq \langle x, \xi \rangle - f^o(x) = u^o(\xi),$$

where the equality is due to $\xi \in \partial f^o(x)$. Since $u_h^o \leq u^o$, it forces $u_h^o(\xi) = u^o(\xi)$.

(b) We show that W_h is a star set, i.e. if $\xi \in W_h$ then $t\xi \in W_h$ for any $t \in [0, 1)$.

Suppose $\xi \in \partial f^o(x)$ with $f^o(x) \leq h$. Take any $x_t \in \partial u^o(t\xi)$, since $t\xi \in \partial f^o(x_t)$, it suffices to show $f^o(x_t) \leq h$. Since $x \in \partial u^o(\xi)$ and $x_t \in \partial u^o(t\xi)$, by the definition of subdifferentials, we have

$$\begin{aligned} u^o(t\xi) &\geq u^o(\xi) + \langle x, t\xi - \xi \rangle , \\ u^o(\xi) &\geq u^o(t\xi) + \langle x_t, \xi - t\xi \rangle . \end{aligned}$$

They imply $\langle x_t, \xi \rangle \leq \langle x, \xi \rangle$. Then we have

$$\begin{aligned} f^o(x_t) &= \langle x_t, t\xi \rangle - u^o(t\xi) \leq t \langle x, \xi \rangle - u^o(t\xi) \\ &\leq t \langle x, \xi \rangle - [u^o(\xi) + \langle x, t\xi - \xi \rangle] \\ &= \langle x, \xi \rangle - u^o(\xi) = f^o(x) \leq h. \end{aligned}$$

Since W_h is a star set, we can describe it as

$$W_h = \{tq \mid q \in \partial P, 0 \leq t \leq r_h(q) \leq 1\} ,$$

where $r_h : \partial P \to [0, 1]$ is a function. Since W_h is closed, $r_h(q)$ is usc in q and nondecreasing in h. We simply write $r_q := r_h(q)$.

(c) We show that $\lim_{h \to \infty} r_h(q) = 1$ uniformly for $q \in \partial P$.

For any $\varepsilon \in (0, 1)$, let $P_\varepsilon := (1 - \varepsilon)P$. By [Fig17, Lemma A.22], $\partial u^o (P_\varepsilon)$ is a compact set in $N_{\mathbb{R}}$. We claim that

$$P_\varepsilon \subset W_h \text{ when } h > H_\varepsilon := \sup_{\partial u^o(P_\varepsilon)} f^o.$$

For any $\xi \in P_\varepsilon$, take $x \in \partial u^o(\xi) \subset \partial u^o (P_\varepsilon)$, we have $\xi \in \partial f^o(x)$ and $f^o(x) \leq H_\varepsilon$, so $\xi \in W_{h > H_\varepsilon}$. By the definition of $r_h(q)$, we have

$$1 - \varepsilon \leq r_h(q) \leq 1, \ \forall q \in \partial P$$

when $h > H_\varepsilon$. Hence **(c)** follows.

(d) For any $q \in \partial P$, u_h^o is affine restricting on the ray $\{tq \mid t \geq r_q := r_h(q)\}$. Explicitly, we have

$$u_h^o(tq) = \frac{t}{r_q}\left(u^o(r_q q) + h\right) - h, \text{ for } t \geq r_q.$$

Proof: For any $q \in \partial P$, fix a $t > r_q$. Since $S_{f^o}(h)$ is compact, the supremum

$$u_h^o(tq) := \sup_{x \in S_{f^o}(h)} \langle x, \xi \rangle - f^o(x) = \sup_{x \in N_{\mathbb{R}}} \langle x, tq \rangle - \left(f^o(x) + I_{S_{f^o}(h)}(x)\right)$$

is attained by some $x^* \in S_{f^o}(h)$, and this occurs if and only if

$$tq \in \partial\left(f^o + I_{S_{f^o}(h)}\right)(x^*) = \partial f^o(x^*) + \partial I_{S_{f^o}(h)}(x^*). \tag{4.7}$$

The last equality is due to Moreau–Rockafellar theorem.

If x^* does not lie on the boundary of $S_{f^o}(h)$, (4.7) implies $tq \in \partial f^o(x^*) \subset W_h$. That is absurd since $t > r_q$. Hence x^* must lie on the boundary and $f^o(x^*) = h$.

By the definition of subdifferentials, we note that

$$\begin{aligned}\partial I_{S_{f^o}(h)}(x^*) &= \{\xi \mid \langle x - x^*, \xi \rangle \leq 0,\ \forall x \in S_{f^o}(h)\} \\ &= \{\lambda\eta \mid \lambda \geq 0,\ \eta \in \partial f^o(x^*)\}.\end{aligned}$$

Thus by (4.7), we can write $tq = \eta_1 + \lambda\eta_2$ with $\eta_i \in \partial f^o(x^*)$ and $\lambda > 0$.

For any $a \geq 1$, we find

$$atq = \eta_1 + (a - 1 + a\lambda)\frac{(a-1)\eta_1 + a\lambda\eta_2}{a - 1 + a\lambda} \in \partial f^o(x^*) + \partial I_{S_{f^o}(h)}(x^*),$$

where the convexity of $\partial f^o(x^*)$ is used. This inclusion relation implies that the supremum

$$u_h^o(atq) := \sup_{x \in N_{\mathbb{R}}} \langle x, atq \rangle - \left(f^o(x) + I_{S_{f^o}(h)}(x)\right)$$

is still attained by x^*. So $u_h^o(atq) = \langle x^*, atq \rangle - h$ for all $a \geq 1$. Since t can be arbitrarily close to r_q, **(d)** follows.

(e) It is ready to show $u = u_h$ for $h \gg 1$. Let

$$v_h := u - u_h = u^o - u_h^o \geq 0.$$

By property **(c)**, for any $q \in \partial P$, the function $[r_q, 1] \ni t \mapsto v_h(tq)$ is convex and vanishes at $t = r_q$ by **(a)**. It implies that this function is nondecreasing, so

$$v_h(tq) \leq v_h(q),\ \forall t \in [r_q, 1]. \tag{4.8}$$

Note that $v_h(q) < \infty$, when q lies in the relative interior of facets, since $u < +\infty$ on P^*.

To estimate the integrals over P, we decompose $P = \bigcup_{\rho \in \mathcal{F}_P(1)} P_\rho$, where P_ρ is the cone with apex 0 and base F_ρ (facet of P associated to ρ). Consider the map

$$[0,1] \times F_\rho \to P_\rho,\ (t, q) \mapsto tq.$$

One can check that the pullback of Lebesgue measure $\mathrm{d}\mu$ along this map is $a_\rho t^{n-1}\mathrm{d}t \wedge \mathrm{d}\sigma$.

By **(a)**, $v_h(tq) = 0$ when $t \in [0, r_q]$. Use (4.8), we have

$$\int_{P_\rho} v_h \mathrm{d}\mu = a_\rho \int_{F_\rho}\int_{r_q}^1 v_h(tq) t^{n-1}\mathrm{d}t\mathrm{d}\sigma \leq \frac{a_\rho}{n}\int_{F_\rho}\left(1 - r_q^n\right) v_h(q)\mathrm{d}\sigma.$$

Take the sum over ρ, we obtain

$$0 \leq \int_P v_h \mathrm{d}\mu \leq \frac{\max_\rho a_\rho}{n}\left(1 - \inf_{q \in \partial P} r_q^n\right)\int_{\partial P} v_h \mathrm{d}\sigma. \tag{4.9}$$

Compare the L^2-norms,

$$\int_P u_h^2 - \int_P u^2 = \int_P (u_h^o + \ell)^2 - \int_P (u^o + \ell)^2 = \int_P (u_h^o)^2 - \int_P (u^o)^2 - 2\int_P v_h \ell$$
$$\leq 2\max_P |\ell| \int_P v_h \leq \frac{2}{n} \max_P |\ell| \max_\rho a_\rho \left(1 - \inf_{q\in\partial P} r_q^n\right) \int_{\partial P} v_h \mathrm{d}\sigma.$$

where $\int_P (u_h^o)^2 \leq \int_P (u^o)^2$ is due to $0 \leq u_h^o \leq u^o$.

We obtain a comparison of functional values,

$$\left(\frac{1}{2}\int_P u_h^2 + \int_{\partial P} u_h\right) - \left(\frac{1}{2}\int_P u^2 + \int_{\partial P} u\right) \leq \left[C\left(1 - \inf_{q\in\partial P} r_q^n\right) - 1\right] \int_{\partial P} v_h \mathrm{d}\sigma.$$

By **(c)**, we know $\inf_{q\in\partial P} r_q^n \to 1$. So there exists $h_0 > 0$, when $h > h_0$, the RHS $\leq -\frac{1}{2}\int_{\partial P} v_h$. Since u is a minimizer of (4.5), we must have $\int_{\partial P} v_h \mathrm{d}\sigma = 0$ for $h > h_0$. By (4.9), it implies $\int_P v_h = 0$, so $u = u_h$ almost everywhere. Since u is continuous on P^* and u_h is continuous everywhere, so $u = u_h$ on P^*. Note u is also lsc, so

$$u_h = u, \text{ on } P \text{ when } h > h_0.$$

Moreover, by (4.6), we have

$$\partial u\,(\mathrm{int}P) = \partial u_h\,(\mathrm{int}P) \subset \partial u_h\,(M_{\mathbb{R}}) = S_{f^o}(h) + p^o$$

when $h > h_0$. Since $S_{f^o}(h)$ is bounded, so is $\partial u\,(\mathrm{int}P)$. □

The following property is due to Li and Zhou. They deal with a general problem and assume a smoothness condition (P2) which is not satisfied by Θ_P, but the proof is still valid.

Theorem 42 ([LZ22, Proposition 3.2])**.** *Convex function $u = -\Theta_P$ satisfies the homogeneous Monge–Ampère equation in the sense of Alexandrov. Precisely, for any compact set $K \subset \mathrm{int}P$, the Lebesgue measure $|\partial u(K)| = 0$.*

5. From shortest GKZ vectors to optimal test-function

Back to the original question that motivated our research, whether maximal Chow-destabilizers quantize maximal K-destabilizers. For toric varieties, we have seen that it reduces to the relation between Θ_P and the asymptotics of shortest GKZ vectors for (P, P_k) as k tends to infinity.

5.1. Quantum functional Ψ_k.

Motivated by the asymptotics of Riemann sums (A.2), we introduce a quantum functional

$$\Psi_k(\eta) := \frac{1}{2k^n}\sum_{u\in P_k} \eta(u)^2 + \frac{2}{k^{n-1}}\left(\sum_{u\in P_k}\eta(u) - k^n\int_P \eta\, \mathrm{d}x\right),\ \eta \in \mathcal{C}(P).$$

By (A.1), we have

$$\Psi_k(\eta) \to \Psi(\eta),\ \forall \eta \in \mathcal{C}(P).$$

Proof of Theorem 4. **Step 1**. Reduce the minimization problem

$$\inf\left\{\Psi_k(\eta) \mid \eta \in \mathcal{C}(P)\right\} \tag{5.1}$$

to (5.3). For any $\eta \in \mathcal{C}(P)$, let $\eta_k = (\eta|_{P_k})^\vee$. By Proposition 17, we have $\eta_k \geq \eta$ and $\eta_k|_{P_k} = \eta|_{P_k}$, these imply $\Psi_k(\eta_k) \leq \Psi_k(\eta)$ with equality only if $\eta_k = \eta$. Hence problem (5.1) can be reduced to

$$\inf\left\{\Psi_k(\eta) \mid \eta \in \mathrm{PL}(P, P_k)\right\}, \tag{5.2}$$

recall $\mathrm{PL}(P, P_k)$ is the image of envelope map $\vee : \mathbb{R}^{P_k} \to \mathcal{C}(P)$.

By Proposition 17 (3), $\vee : \mathcal{C}(P_k) \to \mathrm{PL}(P, P_k)$ is a bijection. Hence we can change the domain of (5.2) to $\mathcal{C}(P_k)$. Since for any $f \in \mathcal{C}(P_k)$, we have $f^\vee|_{P_k} = f$, it follows that

$$\Psi_k\left(f^\vee\right) = \frac{1}{2k^n}\sum_{u\in P_k} f(u)^2 + \frac{2}{k^{n-1}}\left(\sum_{u\in P_k} f(u) - k^n\int_P f^\vee\right) =: \breve{M}_{P_k}(f).$$

By changing the domain, problem (5.2) is equivalent to

$$\inf\left\{\breve{M}_{P_k}(f) \mid f \in \mathcal{C}(P_k)\right\}. \tag{5.3}$$

Step 2. We claim that

$$f^* := 2V_P k^{n+1}\left(\sigma_k - \frac{1}{V_P k^n}\right)$$

is actually the unique minimizer of $\breve{M}_{P_k}$ over larger space $\mathbb{R}^{P_k}$. The uniqueness is obvious, since $\breve{M}_{P_k}$ is strictly convex.

Once this claim holds, since $f^* \in \mathcal{C}(P_k)$ by Theorem 34 and Proposition 17 (2), it is also the unique minimizer of (5.3). Then trace back the reductions in Step 1, we see $(f^*)^\vee$ is the unique minimizer of Ψ_k, that is (1.4).

Next we verify the above claim. For any $f \in \mathbb{R}^{P_k}$, by (3.10), we have

$$\frac{1}{V_P}\int_P f^\vee = \min_{g\in\Sigma(P_k)}\langle f, g\rangle \le \langle f, \sigma_k\rangle = \left\langle f, \sigma_k - \frac{1}{V_P k^n}\right\rangle + \frac{1}{V_P k^n}\sum_{u\in P_k} f(u),$$

$$\left\langle f, \sigma_k - \frac{1}{V_P k^n}\right\rangle \le \|f\|\left\|\sigma_k - \frac{1}{V_P k^n}\right\| \le \frac{1}{4V_P k^{n+1}}\|f\|^2 + V_P k^{n+1}\left\|\sigma_k - \frac{1}{V_P k^n}\right\|^2.$$

Combine these inequalities together, we obtain

$$\breve{M}_{P_k}(f) \ge -2V_P^2 k^{n+2}\left\|\sigma_k - \frac{1}{V_P k^n}\right\|^2.$$

Next we check that this lower bound is achieved by f^*. Denote $f^* = a(\sigma_k - c)$ for short, where $a = 2V_P k^{n+1}$ and $c = \frac{1}{V_P k^n}$. Using identities

$$\sum_{u\in P_k}\sigma_k(u) = 1 \text{ and } \int_P \sigma_k^\vee = V_P\|\sigma_k\|^2,$$

we compute

$$\sum_{u\in P_k} f^*(u) - k^n\int_P (f^*)^\vee = a\left(2 - c\,|P_k| - c^{-1}\|\sigma_k\|^2\right) = -\frac{a}{c}\|\sigma_k - c\|^2.$$

Hence

$$\breve{M}_{P_k}(f^*) = \frac{a^2}{2k^n}\|\sigma_k - c\|^2 - \frac{2}{k^{n-1}}\frac{a}{c}\|\sigma_k - c\|^2 = -2V_P^2 k^{n+2}\|\sigma_k - c\|^2.$$

□

5.2. **Convergence problem.** For $f \in \mathcal{C}(P)$, set

$$\|f\|_{k,2}^2 := \frac{1}{k^n}\sum_{u\in P_k} f(u)^2, \qquad B_k(f) := \frac{2}{k^{n-1}}\left(\sum_{u\in P_k} f(u) - k^n\int_P f\,\mathrm{d}x\right).$$

Thus $\Psi_k(f) = \frac{1}{2}\|f\|_{k,2}^2 + B_k(f)$. We use the following standard compactness form of the convex quadrature lemma. Notice that no control of the values on faces of codimension at least two is assumed.

Lemma 43 (convex quadrature compactness)**.** *Assume that P is Delzant. There is a constant C_P such that*

$$B_k(f) \geq -C_P \|f\|_{k,2}\,, \qquad \|f\|_{L^2(P)} \leq C_P \|f\|_{k,2} \tag{5.4}$$

for every sufficiently large k and every $f \in \mathcal{C}(P)$. Since only the limit $k \to \infty$ is used below, the finitely many smaller values of k are irrelevant. Moreover, suppose that $f_k \in \mathcal{C}(P)$ and $\sup_k \|f_k\|_{k,2} < \infty$. After passing to a subsequence, there is a convex $f \in L^2(P)$ such that $f_k \rightharpoonup f$ weakly in $L^2(P)$ and $f_k \to f$ locally uniformly on $\operatorname{int}P$. We use the lower semicontinuous extension of f to P and allow its boundary integral to be $+\infty$. For this subsequence,

$$\liminf_{k\to\infty} \|f_k\|_{k,2}^2 \geq \|f\|_{L^2(P)}^2\,, \qquad \liminf_{k\to\infty} B_k(f_k) \geq \int_{\partial P} f \,\mathrm{d}\sigma. \tag{5.5}$$

If, in addition,

$$\lim_{k\to\infty} \|f_k\|_{k,2}^2 = \|f\|_{L^2(P)}^2\,, \tag{5.6}$$

then $f_k \to f$ strongly in $L^2(P)$.

Proof. We divide the proof into four steps. Write the facets as $F_1, \ldots, F_N$, choose primitive integral affine functions l_r such that

$$P = \{l_r \geq 0,\ 1 \leq r \leq N\}, \qquad F_r = \{l_r = 0\},$$

and, for $\delta > 0$, put

$$P^\delta := \{x \in P \mid l_r(x) \geq \delta \text{ for every } r\}.$$

Changing the Euclidean norm only changes constants, so below all constants depend only on P. The Delzant assumption is used only in the lattice trapezoid calculation below.

Step 1: the two coercive estimates. Fix a triangulation $\mathcal{T}$ of P whose vertices are lattice points. The standard k-fold edgewise subdivision gives a triangulation $\mathcal{T}_k$ with the following uniform properties:

(1) every vertex of $\mathcal{T}_k$ belongs to P_k;
(2) every simplex has diameter at most C/k;
(3) only finitely many affine shapes occur after rescaling by k;
(4) every vertex belongs to at most C simplices.

If $\Delta \in \mathcal{T}_k$ has vertices $u_0, \ldots, u_n$ and $x = \sum_{i=0}^n \lambda_i u_i$, convexity gives

$$h(x) \leq \sum_{i=0}^n \lambda_i h(u_i). \tag{5.7}$$

Consequently,

$$h_+(x)^2 \leq \left(\sum_{i=0}^n \lambda_i |h(u_i)|\right)^2 \leq \sum_{i=0}^n \lambda_i h(u_i)^2.$$

Integration over Δ, followed by summation over $\mathcal{T}_k$, yields

$$\int_P h_+^2 \,\mathrm{d}x \leq \frac{C}{k^n} \sum_{u\in P_k} h(u)^2. \tag{5.8}$$

The same argument has the following localized form. If $g \geq 0$ is convex and $E \subset P$ is measurable, let E_k be the union of the simplices of $\mathcal{T}_k$ which meet E. Every vertex of E_k is at distance at most C/k from E. Applying the preceding estimate on these simplices gives

$$\int_E g^2 \,\mathrm{d}x \leq \frac{C}{k^n} \sum_{\substack{u\in P_k \\ \operatorname{dist}(u,E)\leq C/k}} g(u)^2. \tag{5.9}$$

We now reduce a general convex function to this nonnegative case. Fix $p_0 \in \operatorname{int} P$ and $r > 0$ such that $B(p_0, 5r) \Subset P$. For every sufficiently large k, choose $p_k \in P_k$ with $|p_k - p_0| \le C/k$, and put

$$Z_k = \{z \in \mathbb{Z}^n : |z| \le rk\}.$$

Then $|Z_k| \ge ck^n$, and all the points $p_k \pm z/k$ and $p_k + 2z/k$, $z \in Z_k$, belong to P_k. Convexity gives, for every $z \in Z_k$,

$$h(p_k) \le \frac{h(p_k + z/k) + h(p_k - z/k)}{2}, \qquad h(p_k) \ge 2h(p_k + z/k) - h(p_k + 2z/k).$$

Average these inequalities over Z_k. For $j \in \{-1, 1, 2\}$, Cauchy–Schwarz and the injectivity of $z \mapsto p_k + jz/k$ give

$$\frac{1}{|Z_k|} \sum_{z \in Z_k} |h(p_k + jz/k)| \le |Z_k|^{-1/2} \left(\sum_{u \in P_k} h(u)^2 \right)^{1/2} \le C \|h\|_{k,2}.$$

It follows that

$$|h(p_k)| \le C \|h\|_{k,2}. \tag{5.10}$$

Choose $q_k \in \partial h(p_k)$. If $q_k \neq 0$, set $e_k = q_k/|q_k|$ and consider

$$S_k = P_k \cap B(p_k + 3re_k, r).$$

For all sufficiently large k, $B(p_k + 3re_k, r) \subset B(p_0, 5r) \subset P$. Hence S_k consists of all the $k^{-1}\mathbb{Z}^n$-points in that ball. A fixed ball contains at least ck^n such points, so $|S_k| \ge ck^n$, uniformly in the direction e_k. For $u \in S_k$, the subgradient inequality and (5.10) give

$$h(u) \ge h(p_k) + \langle q_k, u - p_k \rangle \ge -C\|h\|_{k,2} + 2r|q_k|.$$

Let C_0 denote the constant in the last display. If $|q_k| \le C_0 r^{-1} \|h\|_{k,2}$, the required estimate already holds. Otherwise $h(u) \ge r|q_k|$ for every $u \in S_k$, and therefore

$$k^n \|h\|_{k,2}^2 = \sum_{u \in P_k} h(u)^2 \ge |S_k| r^2 |q_k|^2 \ge ck^n r^2 |q_k|^2.$$

Thus in both cases $|q_k| \le C\|h\|_{k,2}$. The supporting affine function

$$\ell_k(x) = h(p_k) + \langle q_k, x - p_k \rangle$$

and the nonnegative convex function $g_k = h - \ell_k$ consequently satisfy

$$\|\ell_k\|_{C^0(P)} + \|\ell_k\|_{k,2} + \|g_k\|_{k,2} \le C\|h\|_{k,2}. \tag{5.11}$$

Indeed, the bound for $h(p_k)$, the bound for q_k, and the boundedness of P control $\|\ell_k\|_{C^0(P)}$. Then $|P_k| \le Ck^n$ controls $\|\ell_k\|_{k,2}$, and the discrete triangle inequality controls $\|g_k\|_{k,2}$.

Since $g_k \ge 0$, (5.8) applies to g_k itself. Combining it with (5.11) gives

$$\|h\|_{L^2(P)} \le \|g_k\|_{L^2(P)} + \|\ell_k\|_{L^2(P)} \le C\big(\|g_k\|_{k,2} + \|\ell_k\|_{C^0(P)}\big) \le C\|h\|_{k,2}.$$

Thus

$$\|h\|_{L^2(P)} \le C\, \|h\|_{k,2}\,. \tag{5.12}$$

We shall also use the following interior estimate. If $K \Subset K' \Subset \operatorname{int} P$ and h is convex, then

$$\sup_K |h| + \operatorname{Lip}_K(h) \le C_{K,K'}\, \|h\|_{L^2(K')}\,. \tag{5.13}$$

For completeness, choose $r > 0$ so that every ball of radius $4r$ centered in K is contained in K'. Jensen's inequality on $B(x, r)$ gives

$$h(x) \le |B(x,r)|^{-1} \int_{B(x,r)} h \le C_r\, \|h\|_{L^2(K')}\,.$$

This is the required upper bound. For the lower bound, write $h(x) = -M$ and denote the preceding upper bound on the $2r$-neighborhood of K by A. For $z \in B(x,r)$, convexity gives

$$h\left(\frac{x+z}{2}\right) \le \frac{h(x)+h(z)}{2} \le \frac{-M+A}{2}.$$

If $M > 2A$, then $|h| \ge M/4$ on the ball $B(x, r/2)$ obtained as z varies, and therefore $M \le C_r \|h\|_{L^2(K')}$. If $M \le 2A$, the same conclusion is immediate. Thus $|h|$ is bounded on the $2r$-neighborhood of K. Finally, if $q \in \partial h(x)$ and $x \in K$, the subgradient inequalities at $x \pm re_i$ give

$$h(x) \pm r\langle q, e_i\rangle \le h(x \pm re_i).$$

The C^0 bound therefore bounds all components of q, and the subgradient inequality on line segments gives the Lipschitz estimate in (5.13).

We now give the lattice trapezoid calculation which will be used both here and in Step 2. If compact polytopes $K_F \Subset \operatorname{relint} F$ are prescribed, then there are fixed compact convex sets $Q \Subset Q' \Subset \operatorname{int} P$ such that every nonnegative convex g satisfies

$$\sum_{u\in P_k} g(u) - k^n \int_P g\,\mathrm{d}x \ge \frac{1}{2}\sum_F \sum_{u\in K_F\cap P_k} g(u) - Ck^{n-2}\left(\sup_Q g + \operatorname{Lip}_Q(g)\right). \tag{5.14}$$

When $n = 1$, the last term is absent.

Here is the cell bookkeeping behind this estimate. At a codimension r face of a Delzant polytope, the primitive defining functions of the r facets through that face extend to integral affine coordinates. In these coordinates a sufficiently small face neighborhood is an orthant, and the coordinate change has determinant ± 1. Thus it preserves both the lattice and the volume form. Subdivide the neighborhood into half-open lattice boxes of side $1/k$. If R is one such box, with vertices v_ε, $\varepsilon \in \{0,1\}^n$, every $x \in R$ is the convex combination of these vertices with the multilinear barycentric weights. More explicitly, in the box coordinates put $t_i = k(x_i - a_i) \in [0,1]$ and $\lambda_\varepsilon(x) = \prod_i t_i^{\varepsilon_i}(1-t_i)^{1-\varepsilon_i}$. Then $\int_R \lambda_\varepsilon \,\mathrm{d}x = 2^{-n}k^{-n}$. Integrating the convexity inequality gives

$$k^n \int_R g\,\mathrm{d}x \le 2^{-n} \sum_{\varepsilon\in\{0,1\}^n} g(v_\varepsilon). \tag{5.15}$$

After summing (5.15), an interior mesh point occurs in 2^n boxes and has total weight 1; a point in the relative interior of a facet occurs in 2^{n-1} boxes and has weight $1/2$; more generally, a point in the relative interior of a codimension r face has weight 2^{-r}. Each of these weights is no larger than the coefficient 1 with which that point occurs in $\sum_{u\in P_k} g(u)$.

For clarity, the global cell decomposition is obtained as follows. First make the box decomposition on every face and extend it successively in the inward normal directions. At level k, choose each artificial separating interface on the nearest lattice layer and triangulate the one-layer slab between two different box charts. Only finitely many rescaled slab shapes occur. Assign one half of the lattice points on such an interface to each adjacent chart. For a constant function their two half-weights add exactly to the interior weight one. For a general g, subtract its value at one vertex of each slab cell. The constant parts still cancel, whereas the remaining part is bounded by $Ck^{-1}\operatorname{Lip}_Q(g)$ per cell. Since there are $O(k^{n-1})$ cells along all codimension-one interfaces, their total contribution is at most $Ck^{n-2}\operatorname{Lip}_Q(g)$.

Fill the remaining core by lattice boxes and triangulate the finitely many cut-cell shapes. A cut cell which is not already included in a codimension-one slab must meet two separating interfaces, so there are only $O(k^{n-2})$ such cells. All their vertices lie in one fixed compact set $Q \Subset \operatorname{int} P$, and their rescaled volumes and numbers of vertices are uniformly bounded. Their total error is therefore at most $Ck^{n-2}\sup_Q g$. The genuine boundary coefficients are nonnegative, and on each prescribed K_F they equal $1/2$ for all sufficiently large k. Dropping all remaining nonnegative coefficients proves (5.14).

Return to the decomposition $h = g_k + \ell_k$ constructed in (5.11). Take all K_F empty in (5.14) and multiply that inequality by $2/k^{n-1}$. By (5.13), (5.8), and (5.11),

$$\sup_Q g_k + \mathrm{Lip}_Q(g_k) \le C\|g_k\|_{L^2(Q')} \le C\|g_k\|_{k,2} \le C\|h\|_{k,2}.$$

Consequently,

$$B_k(g_k) \ge -\frac{C}{k}\|h\|_{k,2} \ge -C\|h\|_{k,2}. \tag{5.16}$$

For an affine function ℓ_k, the first two coefficients of the weighted Ehrhart polynomial give

$$B_k(\ell_k) = \int_{\partial P} \ell_k \, \mathrm{d}\sigma + \mathrm{O}(k^{-1}) \, \|\ell_k\|_{C^0(P)} \, .$$

Since $|\int_{\partial P} \ell_k \, \mathrm{d}\sigma| \le C\|\ell_k\|_{C^0(P)}$, (5.11) implies $B_k(\ell_k) \ge -C\|h\|_{k,2}$. The linearity of B_k and (5.16) therefore give

$$B_k(h) \ge -C \, \|h\|_{k,2} \, .$$

Together with (5.12), this proves (5.4).

Step 2: the boundary liminf. We first establish the compactness needed in this step. By (5.12), the sequence $\{f_k\}$ is bounded in $L^2(P)$, so a subsequence converges weakly to some $f \in L^2(P)$. Convexity passes to the weak limit: for every nonnegative $\chi \in C_c^\infty(\mathrm{int}P)$ and every vector ξ,

$$\int_P f_k \, \partial_{\xi\xi}\chi \, \mathrm{d}x \ge 0.$$

Weak convergence permits passage to the limit. Thus the distributional Hessian of f is nonnegative, and the standard distributional characterization of convexity gives a convex representative of f, which we continue to denote by f.

For each $K \Subset K' \Subset \mathrm{int}P$, the right-hand side of (5.13) is uniformly bounded. Hence $\{f_k\}$ is uniformly bounded and equicontinuous on K. Choose an exhaustion $K_1 \Subset K_2 \Subset \cdots \Subset \mathrm{int}P$. Arzelà–Ascoli first gives a uniformly convergent subsequence on K_1; from it choose one converging on K_2, and continue inductively. The diagonal subsequence converges locally uniformly on $\mathrm{int}P$. Local uniform convergence implies distributional convergence, while the same subsequence converges weakly in L^2; therefore its local uniform limit is the weak limit f.

Fix compact polytopes $K_F \Subset \mathrm{relint}\, F$, one for each facet. Applying (5.14) to a nonnegative convex sequence g_k and recalling the definition of B_k gives the explicit estimate

$$B_k(g_k) \ge \sum_F \frac{1}{k^{n-1}} \sum_{y \in K_F \cap P_k} g_k(y) - \frac{C}{k}\left(\sup_Q g_k + \mathrm{Lip}_Q(g_k)\right). \tag{5.17}$$

The coefficient is 1 because the facet weight $1/2$ in (5.14) is multiplied by the factor 2 in B_k. The error is $O(k^{-1})$ whenever g_k is locally uniformly bounded on $\mathrm{int}P$, because (5.13) also gives a uniform Lipschitz bound on Q.

We apply this to the subsequence just constructed. Fix a point $p \in \mathrm{int}P$ and choose $q_k \in \partial f_k(p)$. Local uniform boundedness on a ball around p implies that $\{q_k\}$ is bounded: if $B(p, 2r) \Subset P$, the subgradient inequality at $p \pm re_i$ gives

$$|\langle q_k, e_i\rangle| \le r^{-1}\left(|f_k(p + re_i)| + |f_k(p - re_i)| + 2|f_k(p)|\right).$$

After passing to a further subsequence, the supporting affine functions

$$\ell_k(x) = f_k(p) + \langle q_k, x - p\rangle$$

converge uniformly to an affine function ℓ, and $g_k = f_k - \ell_k \geq 0$ converges locally uniformly to $g = f - \ell$ on $\mathrm{int}P$.

For every facet F, primitivity of l_F gives an integral vector v_F with $l_F(v_F) = 1$. Since K_F is separated from all the other facets, this vector points into P above K_F for a sufficiently short time. Thus we may choose $\delta > 0$ so small that

$$\{y + tv_F \mid y \in K_F,\ 0 \leq t \leq 2\delta\} \Subset P \setminus \bigcup_{F' \neq F} F'.$$

Put $m_k = \lfloor k\delta \rfloor$. Then $y + jv_F/k \in P_k$ for $y \in K_F \cap P_k$ and $0 \leq j \leq 2m_k$, once k is sufficiently large.

For $y \in K_F \cap k^{-1}\mathbb{Z}^n$, convexity on the segment $[y, y + 2m_k v_F/k]$ gives

$$g_k(y) \geq 2g_k(y + m_k v_F/k) - g_k(y + 2m_k v_F/k). \tag{5.18}$$

Both points on the right stay in a compact subset of $\mathrm{int}P$. Hence local uniform convergence and the ordinary Riemann-sum theorem on F imply, using (5.17) and the boundedness of $\sup_Q g_k + \mathrm{Lip}_Q(g_k)$,

$$\liminf_{k\to\infty} B_k(g_k) \geq \sum_F \int_{K_F} (2g(y + \delta v_F) - g(y + 2\delta v_F))\, \mathrm{d}\sigma_F.$$

For a one-dimensional convex function $t \mapsto g(y + tv_F)$, the quantities $2g(y+\delta v_F) - g(y + 2\delta v_F)$ converge increasingly along dyadic $\delta \downarrow 0$ to the lower semicontinuous boundary value $g(y)$. Indeed, if $\phi(t) = g(y + tv_F)$, monotonicity of the secant slopes gives

$$2\big(\phi(t) - \phi(t/2)\big) \leq \phi(2t) - \phi(t),$$

which is exactly $2\phi(t/2) - \phi(t) \geq 2\phi(t) - \phi(2t)$. The limit is $\phi(0)$ by the definition of the lower semicontinuous extension. On a fixed K_F these functions have a common lower bound after restricting to $0 < \delta \leq \delta_0$, because their possible negative part is bounded by $g(y + 2\delta v_F)$ on a compact subset of the interior. We may therefore add one constant and apply monotone convergence. We first let dyadic $\delta \downarrow 0$ and then take an increasing exhaustion $K_F \nearrow \mathrm{relint}F$. A second application of monotone convergence gives

$$\liminf_{k\to\infty} B_k(g_k) \geq \sum_F \int_F g\, \mathrm{d}\sigma_F.$$

Finally, the affine Ehrhart formula and $\ell_k \to \ell$ uniformly give $B_k(\ell_k) \to \int_{\partial P} \ell\, \mathrm{d}\sigma$. Since $f_k = g_k + \ell_k$, we obtain

$$\liminf_{k\to\infty} B_k(f_k) \geq \int_{\partial P} f\, \mathrm{d}\sigma. \tag{5.19}$$

If the right-hand side is $+\infty$, the same argument applied to an increasing exhaustion by the K_F shows that the left-hand side is $+\infty$ as well.

Step 3: the quadratic liminf. Fix $\delta > 0$ for which P^δ has nonempty interior. Local uniform convergence and the ordinary Riemann-sum theorem give

$$\lim_{k\to\infty} \frac{1}{k^n} \sum_{u \in P_k \cap P^\delta} f_k(u)^2 = \int_{P^\delta} f^2\, \mathrm{d}x.$$

All omitted summands are nonnegative, hence

$$\liminf_{k\to\infty} \|f_k\|_{k,2}^2 \geq \int_{P^\delta} f^2\, \mathrm{d}x.$$

Letting $\delta \downarrow 0$ and using monotone convergence proves

$$\liminf_{k\to\infty} \|f_k\|_{k,2}^2 \geq \|f\|_{L^2(P)}^2 .$$

Together with (5.19), this proves (5.5).

Step 4: absence of an L^2 boundary defect. Assume now (5.6). For every fixed $\delta > 0$, the interior Riemann-sum convergence then implies

$$\begin{aligned}\limsup_{k\to\infty} \frac{1}{k^n} \sum_{u\in P_k\setminus P^\delta} f_k(u)^2 &\le \|f\|^2_{L^2(P)} - \int_{P^\delta} f^2\,\mathrm{d}x \\ &= \int_{P\setminus P^\delta} f^2\,\mathrm{d}x. \end{aligned} \tag{5.20}$$

For each k, apply (5.11) to f_k and write

$$f_k = g_k + \ell_k, \qquad g_k \ge 0, \qquad \|\ell_k\|_{C^0(P)} \le C\|f_k\|_{k,2} \le C.$$

Set $E = P \setminus P^{2\delta}$ and

$$U_{k,\delta} = \{u \in P_k : \mathrm{dist}(u, E) \le C/k\}.$$

For all sufficiently large k, one has $U_{k,\delta} \subset P_k \setminus P^\delta$. Moreover, the boundary strip E has volume at most $C\delta$, and $|U_{k,\delta}| \le C(k^n\delta + k^{n-1})$. Using (5.9) for g_k, followed by $g_k^2 \le 2f_k^2 + 2\ell_k^2$, gives

$$\begin{aligned}\int_E f_k^2\,\mathrm{d}x &\le 2\int_E g_k^2\,\mathrm{d}x + 2|E|\|\ell_k\|^2_{C^0(P)} \\ &\le \frac{C}{k^n}\sum_{u\in U_{k,\delta}} f_k(u)^2 + C\left(k^{-n}|U_{k,\delta}| + |E|\right)\|\ell_k\|^2_{C^0(P)}.\end{aligned}$$

For fixed δ, the last coefficient is at most $C\delta$ once k is sufficiently large. Hence (5.20) implies

$$\limsup_{k\to\infty}\int_{P\setminus P^{2\delta}} f_k^2\,\mathrm{d}x \le C\int_{P\setminus P^\delta} f^2\,\mathrm{d}x + C\delta.$$

On $P^{2\delta}$ the convergence is uniform, hence strong in L^2. Therefore

$$\begin{aligned}\limsup_{k\to\infty}\|f_k - f\|^2_{L^2(P)} &\le 2\limsup_{k\to\infty}\int_{P\setminus P^{2\delta}} f_k^2\,\mathrm{d}x + 2\int_{P\setminus P^{2\delta}} f^2\,\mathrm{d}x \\ &\le 2C\int_{P\setminus P^\delta} f^2\,\mathrm{d}x + 2C\delta + 2\int_{P\setminus P^{2\delta}} f^2\,\mathrm{d}x.\end{aligned}$$

All terms tend to zero as $\delta \downarrow 0$, because $f \in L^2(P)$. Hence $f_k \to f$ strongly in $L^2(P)$, which proves the last assertion. □

Remark 44. The Delzant hypothesis in Lemma 43 cannot be omitted for a general lattice polytope. For example, let

$$P_m = \mathrm{conv}\{(0,0),(1,0),(1,m)\}, \qquad m > 6,$$

and write a point as (x_1, x_2). The primitive edge directions at the origin have determinant m, so P_m is not Delzant. The convex functions

$$g_k(x_1,x_2) = k\max\{1 - kx_1, 0\}$$

vanish at every point of $(P_m)_k$ except the origin, where their value is k. Therefore

$$\|g_k\|^2_{k,2} = 1, \qquad \int_{P_m} g_k\,\mathrm{d}x = \int_0^{1/k} kmx_1(1-kx_1)\,\mathrm{d}x_1 = \frac{m}{6k},$$

and hence

$$B_k(g_k) = \frac{2}{k}\left(k - k^2\frac{m}{6k}\right) = 2\left(1 - \frac{m}{6}\right) < 0.$$

On the other hand, $g_k \to 0$ locally uniformly away from the origin and $g_k \rightharpoonup 0$ weakly in $L^2(P_m)$; indeed $\int_{P_m} g_k^2 \,\mathrm{d}x = m/12$. Thus the boundary liminf in (5.5) would read $2(1-m/6) \geq 0$, which is false. The lattice-box weights in (5.15) explain exactly why smoothness of the transverse lattice cones removes this defect.

Proof of Theorem 5. Put $\psi = -\Theta_P$. By Proposition 40 and Theorem 1, $\psi \in \mathcal{C}(P)$ is the unique minimizer of Ψ. Minimality of ψ_k and (A.2) give

$$\limsup_{k\to\infty} \Psi_k(\psi_k) \leq \lim_{k\to\infty} \Psi_k(\psi) = \Psi(\psi) = \inf \Psi. \tag{5.21}$$

On the other hand, (5.4) and Young's inequality imply

$$\Psi_k(f) \geq \frac{1}{2}\|f\|_{k,2}^2 - C_P\|f\|_{k,2} \geq \frac{1}{4}\|f\|_{k,2}^2 - C_P^2.$$

Together with (5.21), this shows that $\sup_k \|\psi_k\|_{k,2} < \infty$.

Step 1. We first prove convergence of the discrete quadratic terms. Set

$$a_k := \|\psi_k\|_{k,2}^2 .$$

The preceding coercivity argument shows that $\{a_k\}$ is bounded. Let A be any subsequential limit of $\{a_k\}$, and choose a subsequence $\{k_j\}$ such that

$$\lim_{j\to\infty} a_{k_j} = A. \tag{5.22}$$

Lemma 43, applied to $\{\psi_{k_j}\}$, provides a further subsequence, still denoted by $\{k_j\}$, and a convex function $u \in L^2(P)$ such that

$$\psi_{k_j} \rightharpoonup u \quad \text{weakly in } L^2(P), \qquad \psi_{k_j} \longrightarrow u \quad \text{locally uniformly on } \mathrm{int}P.$$

We claim first that $\int_{\partial P} u \,\mathrm{d}\sigma$ is finite. Indeed, Lemma 43 interprets this integral in the extended sense. If it were $+\infty$, then (5.22) and (5.5) would imply

$$\liminf_{j\to\infty} \Psi_{k_j}(\psi_{k_j}) \geq \frac{1}{2}A + \int_{\partial P} u \,\mathrm{d}\sigma = +\infty,$$

contradicting (5.21). Thus $u \in \mathcal{C}_1(P) \cap L^2(P)$ and $\Psi(u)$ is well-defined.

Since $a_{k_j} \to A$, the elementary inequality $\liminf(x_j + y_j) \geq \lim x_j + \liminf y_j$ and (5.5) give

$$\begin{aligned}
\liminf_{j\to\infty} \Psi_{k_j}(\psi_{k_j}) &= \liminf_{j\to\infty} \left(\frac{1}{2}a_{k_j} + B_{k_j}(\psi_{k_j})\right) \\
&\geq \frac{1}{2}A + \liminf_{j\to\infty} B_{k_j}(\psi_{k_j}) \\
&\geq \frac{1}{2}A + \int_{\partial P} u \,\mathrm{d}\sigma \\
&\geq \frac{1}{2}\|u\|_{L^2(P)}^2 + \int_{\partial P} u \,\mathrm{d}\sigma = \Psi(u).
\end{aligned}$$

Here the last inequality uses $A = \lim_j a_{k_j} \geq \|u\|_{L^2(P)}^2$, which is the first inequality in (5.5). Combining this with (5.21) and the minimality of ψ gives the complete chain

$$\begin{aligned}
\Psi(\psi) &\geq \limsup_{j\to\infty} \Psi_{k_j}(\psi_{k_j}) \geq \liminf_{j\to\infty} \Psi_{k_j}(\psi_{k_j}) \\
&\geq \frac{1}{2}A + \int_{\partial P} u \,\mathrm{d}\sigma \geq \Psi(u) \geq \Psi(\psi).
\end{aligned}$$

Hence every inequality in this chain is an equality. In particular, $\Psi(u) = \Psi(\psi)$, so the uniqueness of the minimizer of Ψ implies

$$u = \psi. \tag{5.23}$$

Moreover, the equality chain contains the identity

$$\frac{1}{2}A + \int_{\partial P} u \, \mathrm{d}\sigma = \Psi(u) = \frac{1}{2} \|u\|^2_{L^2(P)} + \int_{\partial P} u \, \mathrm{d}\sigma.$$

Since the boundary integral is finite, subtracting it from both sides gives

$$A = \|u\|^2_{L^2(P)} = \|\psi\|^2_{L^2(P)} .$$

Thus every subsequential limit of the bounded sequence $\{a_k\}$ equals $\|\psi\|^2_{L^2(P)}$. It follows that the whole sequence converges:

$$\lim_{k\to\infty} \|\psi_k\|^2_{k,2} = \|\psi\|^2_{L^2(P)} . \tag{5.24}$$

For completeness, if this conclusion failed, there would be an $\varepsilon > 0$ and a subsequence along which $|a_k - \|\psi\|^2_{L^2(P)}| \geq \varepsilon$. Boundedness would give a further convergent subsequence, whose limit would be a subsequential limit different from $\|\psi\|^2_{L^2(P)}$, a contradiction.

Step 2. We next prove strong L^2 convergence of the functions. Let $\{\psi_{k_j}\}$ be an arbitrary subsequence. By Lemma 43, after passing to a further subsequence $\{\psi_{k_{j_\ell}}\}$, there is a convex $v \in L^2(P)$ such that

$$\psi_{k_{j_\ell}} \rightharpoonup v \quad \text{weakly in } L^2(P)$$

and the convergence is locally uniform on $\mathrm{int}P$. The boundary integral of v must be finite: otherwise the boundary liminf in (5.5) would contradict (5.21), exactly as in Step 1. Hence $v \in \mathcal{C}_1(P) \cap L^2(P)$.

By (5.24),

$$\lim_{\ell\to\infty} \left\|\psi_{k_{j_\ell}}\right\|^2_{k_{j_\ell},2} = \|\psi\|^2_{L^2(P)} .$$

The quadratic liminf in (5.5) first shows that $\|\psi\|^2_{L^2(P)} \geq \|v\|^2_{L^2(P)}$. Using this inequality, the boundary liminf, (5.21), and minimality of ψ, we obtain

$$\begin{aligned}
\Psi(\psi) &\geq \liminf_{\ell\to\infty} \Psi_{k_{j_\ell}}(\psi_{k_{j_\ell}}) \\
&\geq \frac{1}{2} \|\psi\|^2_{L^2(P)} + \int_{\partial P} v \, \mathrm{d}\sigma \\
&\geq \frac{1}{2} \|v\|^2_{L^2(P)} + \int_{\partial P} v \, \mathrm{d}\sigma = \Psi(v) \geq \Psi(\psi).
\end{aligned}$$

Thus equality holds throughout. The uniqueness of the minimizer gives $v = \psi$, and consequently

$$\lim_{\ell\to\infty} \left\|\psi_{k_{j_\ell}}\right\|^2_{k_{j_\ell},2} = \|v\|^2_{L^2(P)} .$$

Therefore the no-defect hypothesis (5.6) in the last assertion of Lemma 43 is satisfied. That assertion yields

$$\psi_{k_{j_\ell}} \longrightarrow \psi \quad \text{strongly in } L^2(P).$$

We have proved that every subsequence of $\{\psi_k\}$ has a further subsequence converging strongly to ψ. This implies convergence of the whole sequence. Indeed, otherwise there would be an $\varepsilon > 0$ and a subsequence $\{\psi_{k_j}\}$ satisfying

$$\left\|\psi_{k_j} - \psi\right\|_{L^2(P)} \geq \varepsilon \quad \text{for every } j,$$

but this subsequence would have a further subsequence converging strongly to ψ, which is impossible. Hence

$$\psi_k \longrightarrow \psi = -\Theta_P \quad \text{strongly in } L^2(P). \tag{5.25}$$

Since $\psi = -\Theta_P$, (5.24) is precisely (1.5). Furthermore, (1.4) and (5.25) give

$$\left\| V_P k^n \sigma_k^\vee - 1 + \frac{\Theta_P}{2k} \right\|_{L^2(P)} = \frac{1}{2k} \|\psi_k + \Theta_P\|_{L^2(P)} = \mathrm{o}(k^{-1}),$$

which proves (1.6).

Step 3. Finally, assume that $\{\psi_k\}$ is C^0-precompact. We show that the convergence is uniform. If not, there would be an $\varepsilon > 0$ and a subsequence $\{\psi_{k_j}\}$ such that

$$\left\| \psi_{k_j} - \psi \right\|_{C^0(P)} \geq \varepsilon \quad \text{for every } j. \tag{5.26}$$

By C^0-precompactness, a further subsequence converges uniformly to some $w \in \mathcal{C}(P)$. Uniform convergence implies convergence to w in $L^2(P)$, whereas (5.25) says that the same subsequence converges to ψ in $L^2(P)$. Uniqueness of the L^2 limit gives $w = \psi$ almost everywhere. Both functions are continuous on P, so $w = \psi$ everywhere. The further subsequence therefore converges uniformly to ψ, contradicting (5.26). Thus $\psi_k \to \psi$ uniformly. Finally,

$$\left\| V_P k^n \sigma_k^\vee - 1 + \frac{\Theta_P}{2k} \right\|_{C^0(P)} = \frac{1}{2k} \|\psi_k + \Theta_P\|_{C^0(P)} = \mathrm{o}(k^{-1}),$$

which is (1.7). □

Remark 45. The theorem gives unconditional L^2 convergence but does not control values on lower-dimensional faces. To obtain uniform convergence, we still need to verify the C^0-precompactness of $\psi_k = 2V_P k^{n+1}\sigma_k^\vee - 2k$.

Step1.

Remaining questions

(1) Show the sequence ψ_k is uniformly bounded and equicontinuous, thereby upgrading the unconditional L^2 convergence in Theorem 5 to uniform convergence.
(2) Since Theorem 38 (4) shows that Θ_P induces a semistable decomposition of P, we should investigate the semistability of each piece of the subdivision induced by σ_A.
(3) Is there a relation between the asymptotics of $\Sigma(P_k)$ and the space $\mathcal{S}(P)$ (4.3) of prescribed scalar curvatures?
(4) An effective algorithm to compute σ_A without knowing the entire secondary polytope.

Appendix A. Two-terms expansion for Riemann sum of convex functions

We establish a folklore expansion of Riemann sum, which is crucial in the proof of Theorem 5.

Theorem 46. *Let $P \subset \mathbb{R}^n$ be a full-dimensional lattice polytope. For any $k \in \mathbb{Z}_{>0}$, set $P_k = P \cap k^{-1}\mathbb{Z}^n$. Let $f_k : P \to \mathbb{R}$ be a sequence of convex functions (continuous up to ∂P) which uniformly converges to f. Then we have*

$$\lim_{k\to\infty} \frac{1}{k^{n-1}} \left(\sum_{u \in P_k} f_k(u) - k^n \int_P f_k \, \mathrm{d}x \right) = \frac{1}{2} \int_{\partial P} f \, \mathrm{d}\sigma. \tag{A.1}$$

In particular, when $f_k = f$ for all k, we have expansion

$$\sum_{u \in P_k} f(u) = k^n \int_P f \, \mathrm{d}x + \frac{k^{n-1}}{2} \int_{\partial P} f \, \mathrm{d}\sigma + \mathrm{o}(k^{n-1}). \tag{A.2}$$

In the above, $\mathrm{d}x$ *is the standard Lebesgue measure, and the boundary measure* $\mathrm{d}\sigma$ *is defined as follows. For each facet* $F \subset \partial P$*, choose* $p_F \in F \cap \mathbb{Z}^n$ *and set* $\Lambda_F = (\mathrm{Aff}(F) - p_F) \cap \mathbb{Z}^n$*. Then* $\mathrm{d}\sigma|_F$ *is the translation-invariant measure on* $\mathrm{Aff}(F)$ *for which the lattice* Λ_F *has covolume one. This definition is independent of the choice of* p_F*.*

Remark 47. (1) when f is smooth on an open set containing P, (A.2) is just the leading two terms of the Euler–Maclaurin expansion. This expansion holds for every lattice polytope; no simplicity or unimodularity assumption, and hence no Delzant assumption, is needed. See [Tat11] and its references. (2) if f is merely continuous, (A.2) may fail even in one dimension (Weierstrass type functions). (3) If f is convex and piece-wisely linear with rational coefficients, (A.2) can be derived from the asymptotics of lattice point counting, see [Don02, Prop. 4.1.3]. In our setting, (A.2) cannot be derived from these results via a direct approximation. The key in the following proof is decomposing the Riemann sum into interior part and boundary part, and we handle the boundary part via approximation.

Proof. **Step 1**. We extend f_k, f to $\mathbb{R}^n$ such that uniform convergence still holds and they share a common continuity modulus $\hat{\omega}$, i.e. for $g = f_k$ or f, we have

$$|g(x_1) - g(x_2)| \le \hat{\omega}\left(\|x_1 - x_2\|\right), \; \forall x_i \in \mathbb{R}^n.$$

We will use the norm $\|x\| = \max_{1 \le i \le n} |x_i|$ and distance $d(x, y) = \|x - y\|$. Let

$$\omega_k(\delta) := \sup\left\{ |f_k(x) - f_k(y)| \mid x, y \in P, \; d(x, y) \le \delta \right\},$$

be the minimal continuity modulus of f_k. By uniform convergence, the family $\{f_k\}$ is equicontinuous on the compact set P. Indeed, the tail of the sequence is uniformly close to the continuous function f, while a finite initial part has a common continuity modulus. Consequently,

$$\omega(\delta) := \sup_k \omega_k(\delta) \to 0, \text{ as } \delta \to 0.$$

The same estimate holds for the uniform limit f. Thus, for $g = f_k$ or f, we have

$$|g(x) - g(y)| \le \omega\left(\|x - y\|\right), \; \forall x, y \in P.$$

To apply McShane's extension, take a nondecreasing concave majorant $\hat{\omega}$ of ω, with $\hat{\omega}(0) = 0$ and $\hat{\omega}(\delta) \to 0$ as $\delta \to 0$. Such a concave modulus exists for every common modulus of continuity. It is subadditive, i.e. $\hat{\omega}(\delta_1 + \delta_2) \le \hat{\omega}(\delta_1) + \hat{\omega}(\delta_2)$. Then we define McShane's extension:

$$\tilde{f}_k(x) := \inf_{y \in P} f_k(y) + \hat{\omega}\left(\|x - y\|\right), \; \forall x \in \mathbb{R}^n,$$

and $\tilde{f}(x)$ is defined similarly. For $x \in P$, the choice $y = x$ gives $\tilde{f}_k(x) \le f_k(x)$, whereas the modulus estimate on P gives the reverse inequality. Thus the extension coincides with the original function on P.

They share a common continuity modulus $\hat{\omega}$. For any $x_1, x_2 \in \mathbb{R}^n$ and $y \in P$, by the subadditivity, we have

$$\tilde{f}_k(x_1) \le f_k(y) + \hat{\omega}\left(\|x_1 - y\|\right) \le f_k(y) + \hat{\omega}\left(\|x_2 - y\|\right) + \hat{\omega}\left(\|x_1 - x_2\|\right).$$

Take $\inf_{y \in P}$ on both sides,

$$\tilde{f}_k(x_1) \le \tilde{f}_k(x_2) + \hat{\omega}\left(\|x_1 - x_2\|\right).$$

Swap x_1, x_2, we obtain

$$\left|\tilde{f}_k(x_1) - \tilde{f}_k(x_2)\right| \leq \hat{\omega}\left(\|x_1 - x_2\|\right).$$

This also holds for $\tilde{f}$.

Finally, we show that $\tilde{f}_k$ uniformly converges to $\tilde{f}$ on $\mathbb{R}^n$. Take any $\varepsilon > 0$, we have $|f_k - f| < \varepsilon$ on P when $k > N$. Then for any $x \in \mathbb{R}^n$ and $y \in P$, we have

$$f(y) + \hat{\omega}\left(\|x-y\|\right) - \varepsilon \leq f_k(y) + \hat{\omega}\left(\|x-y\|\right) \leq f(y) + \hat{\omega}\left(\|x-y\|\right) + \varepsilon.$$

Take $\inf_{y\in P}$ on both sides, we obtain $\left|\tilde{f}_k(x) - \tilde{f}(x)\right| < \varepsilon$ for all $x \in \mathbb{R}^n$ when $k > N$.

We will simply denote $\tilde{f}_k, \tilde{f}$ by f_k and f respectively.

Step 2: interior-boundary decomposition.

For a (scaled) lattice point $u \in k^{-1}\mathbb{Z}^n$, we define a cell (or box)

$$C_k(u) := u + \left(-\frac{1}{2k}, \frac{1}{2k}\right)^n = \left\{x \in \mathbb{R}^n \mid \|x-u\| < \frac{1}{2k}\right\}$$

with volume $|C_k(u)| = k^{-n}$. Let

$$A_k := \left\{u \in k^{-1}\mathbb{Z}^n \mid C_k(u) \subset P\right\},\ B_k := \left\{u \in k^{-1}\mathbb{Z}^n \mid C_k(u) \cap \partial P \neq \emptyset\right\},$$

refer to the figure below. There exists a constant C_P such that

$$|A_k| \leq C_P k^n, \qquad |B_k| \leq C_P k^{n-1} \quad \text{for all } k \geq 1.$$

To see the second estimate, observe that the disjoint cells indexed by B_k are contained in the k^{-1}-neighborhood of ∂P. Since P has finitely many facets, this neighborhood has volume $O(k^{-1})$; multiplying by the reciprocal cell volume k^n gives $|B_k| = O(k^{n-1})$. The first estimate follows similarly from the boundedness of P. We decompose the sum and integral (χ_P is the indicator function),

$$\sum_{u\in P_k} f_k(u) = \sum_{u\in A_k} f_k(u) + \sum_{u\in B_k} f_k(u)\chi_P(u),$$

$$\int_P f_k = \sum_{u\in A_k}\int_{C_k(u)} f_k + \sum_{u\in B_k}\int_{C_k(u)\cap P} f_k. \tag{A.3}$$

We try to find cancellations between them (modulo $\mathrm{o}(k^{n-1})$-errors).

For the integral $\int_{C_k(u)\cap P} f_k$, by the uniform continuity, we have

$$\left|\int_{C_k(u)\cap P} \left(f_k(x) - f_k(u)\right) \mathrm{d}x\right| \leq k^{-n}\hat{\omega}\left(\frac{1}{2k}\right).$$

Thus

$$\sum_{u\in B_k}\int_{C_k(u)\cap P} f_k = \sum_{u\in B_k} f_k(u)\left|C_k(u)\cap P\right| + r_k,$$
$$\text{where } |r_k| \leq |B_k|\, k^{-n}\hat{\omega}\left(\frac{1}{2k}\right) \leq \frac{C_P}{k}\hat{\omega}\left(\frac{1}{2k}\right).$$

In particular, $|k^n r_k| \leq C_P k^{n-1}\hat{\omega}((2k)^{-1}) = \mathrm{o}(k^{n-1})$. Put this back into (A.3), we obtain

$$\sum_{u\in P_k} f_k(u) - k^n\int_P f_k = \sum_{u\in A_k}\left(f_k(u) - k^n\int_{C_k(u)} f_k\right) + \sum_{u\in B_k} f_k(u) w_k(u) + \mathrm{o}\left(k^{n-1}\right)$$
$$\text{define} =: k^{n-1}\mathcal{A}_k(f_k) + k^{n-1}\mathcal{B}_k(f_k) + \mathrm{o}\left(k^{n-1}\right), \tag{A.4}$$

where the weight $w_k(u)$ is

$$w_k(u) := \chi_P(u) - k^n \left|C_k(u) \cap P\right| = \begin{cases} \frac{|C_k(u)\backslash P|}{|C_k(u)|}, & u \in P, \\ -\frac{|C_k(u)\cap P|}{|C_k(u)|}, & u \notin P. \end{cases}$$

Note that $|w_k(u)| \leq 1$. Now, the question is reduced to prove

$$\mathcal{A}_k(f_k) \to 0 \text{ and } \mathcal{B}_k(f_k) \to \frac{1}{2}\int_{\partial P} f \mathrm{d}\sigma.$$

Step 3: show

$$\mathcal{B}_k(f_k) := \frac{1}{k^{n-1}} \sum_{u \in B_k} f_k(u) w_k(u) \to \frac{1}{2}\int_{\partial P} f \mathrm{d}\sigma. \tag{A.5}$$

In this step, the convexity of f_k is not used. We derive (A.5) from the classical smooth Euler–Maclaurin formula via approximation, so it is not a direct proof. Before that, we give an intuitive but not fully rigorous demonstration. Readers may skip this part.

A heuristic proof of (A.5). Let $\partial^{>1}P$ be the union of faces of P with codimension > 1, and we set

$$B_k^{>1} := \left\{ u \in B_k \mid d_\infty\left(u, \partial^{>1}P\right) < \frac{1}{k} \right\}$$

(black cells in the figure). Note that $\left|B_k^{>1}\right| = \mathrm{O}\left(k^{n-2}\right)$, so we have $\sum_{u \in B_k^{>1}} f_k(u) w_k(u) = \mathrm{O}\left(k^{n-2}\right)$, so $B_k^{>1}$ summand makes no contribution to $\lim_k \mathcal{B}_k(f_k)$.

Consider the other summands with $u \in B_k \backslash B_k^{>1}$, u may lie either inside, outside or on the boundary of P. We claim that only the summands that $u \in \partial P$ make contribution to the limit $\lim_k \mathcal{B}_k$. As the figure illustrates, except $\mathrm{O}\left(k^{n-2}\right)$ many points, each $u \in \mathrm{int}P \backslash B_k^{>1}$ can be assigned a partner u' outside of P such that

$$w_k(u) = -w_k(u') \text{ and } d_\infty(u, u') \leq \frac{C}{k}.$$

In the figure, partner points are connected by a line segment. For the associated two-terms in the sum $\mathcal{B}_k(f)$, we have

$$\left|f_k(u) w_k(u) + f_k(u') w_k(u')\right| = \left|f_k(u) - f_k(u')\right| \left|w_k(u)\right| \leq \hat{\omega}\left(\frac{C}{k}\right).$$

On the other hand, for $u \in B_k^\partial := \left(B_k \cap \partial P\right) \backslash B_k^{>1}$, we have $w_k(u) = \frac{1}{2}$. Combining all these estimates, we conclude

$$\lim_{k\to\infty} \mathcal{B}_k(f_k) = \lim_{k\to\infty} \frac{1}{k^{n-1}} \sum_{u \in B_k^\partial} \frac{1}{2} f_k(u) = \frac{1}{2}\int_{\partial P} f \mathrm{d}\sigma.$$

FIGURE A.1. colored boxes are $C_k(u)$ with $u \in B_k$.

Rigorous proof of (A.5). For any $\varepsilon > 0$, take a smooth function g on $\mathbb{R}^n$ such that

$$|f - g| \leq \varepsilon \text{ on } U\left(P, 1\right) := \left\{x \in \mathbb{R}^n \mid d(x, P) < 1\right\}.$$

We will show $\mathcal{A}_k(g) \to 0$ below. The classical smooth Euler–Maclaurin formula [Tat11] gives

$$\sum_{u \in P_k} g(u) - k^n \int_P g \,\mathrm{d}x = \frac{k^{n-1}}{2} \int_{\partial P} g \,\mathrm{d}\sigma + \mathrm{O}_g(k^{n-2}).$$

This two-term formula is valid for an arbitrary lattice polytope: the facets produce the displayed k^{n-1} term, while all faces of codimension at least two contribute only to lower-order terms. In particular, no Delzant assumption is used here. Combining this formula with (A.4) and $\mathcal{A}_k(g) \to 0$ gives $\mathcal{B}_k(g) \to \frac{1}{2}\int_{\partial P} g\mathrm{d}\sigma$. Hence there exists $K > 0$ (depending on g) such that when $k > K$, we have

$$\left|\mathcal{B}_k(g) - \frac{1}{2}\int_{\partial P} g\mathrm{d}\sigma\right| < \varepsilon,\ B_k \subset U(P,1),$$
$$\text{and } |f_k - f| < \varepsilon \text{ on } \mathbb{R}^n.$$

Thus $|f_k - g| \le 2\varepsilon$ on $U(P,1)$. These imply

$$\begin{aligned}\left|\mathcal{B}_k(f_k) - \frac{1}{2}\int_{\partial P} f\mathrm{d}\sigma\right| &\le |\mathcal{B}_k(f_k) - \mathcal{B}_k(g)| + \left|\mathcal{B}_k(g) - \frac{1}{2}\int_{\partial P} g\mathrm{d}\sigma\right| + \frac{1}{2}\int_{\partial P}|g-f|\,\mathrm{d}\sigma \\ &\le \frac{1}{k^{n-1}}\sum_{u\in B_k}|f_k(u) - g(u)| + \varepsilon + \frac{\varepsilon}{2}|\partial P|_{\mathrm{d}\sigma} \\ &\le \frac{2\varepsilon}{k^{n-1}}|B_k| + \varepsilon + \frac{\varepsilon}{2}|\partial P|_{\mathrm{d}\sigma} \le C'_P\varepsilon, \text{ for } k > K.\end{aligned}$$

Thus (A.5) holds.

Now we show $\mathcal{A}_k(g) \to 0$. Since g is smooth, near each $u \in A_k$, we have Taylor expansion

$$g(x) = g(u) + \nabla g(u)\cdot(x-u) + R_u(x),\ |R_u(x)| \le C_g\|x-u\|^2, \tag{A.6}$$

where the constant C_g depends on the C^2-norm of g. Integrating both sides over $C_k(u)$, we obtain

$$\int_{C_k(u)} g(x)\mathrm{d}x = k^{-n}g(u) + \int_{C_k(u)} R_u(x)\mathrm{d}x,$$

where the integral of linear term vanishes by symmetry. Hence

$$\left|g(u) - k^n\int_{C_k(u)} g\right| \le k^n\int_{C_k(u)}|R_u(x)|\,\mathrm{d}x \le C_g k^n\int_{C_k(u)}\|x-u\|^2\,\mathrm{d}x = \mathrm{O}\left(k^{-2}\right).$$

It implies $|\mathcal{A}_k(g)| \le \frac{1}{k^{n-1}}|A_k|\,\mathrm{O}\left(k^{-2}\right) = \mathrm{O}\left(k^{-1}\right) \to 0$.

Step 4: show

$$\mathcal{A}_k(f_k) := \frac{1}{k^{n-1}}\sum_{u\in A_k}\left(f_k(u) - k^n\int_{C_k(u)} f_k\right) \to 0.$$

Because $C_k(u) \subset P$ for every $u \in A_k$, Jensen's inequality can be applied to the original convex function on the whole cell and gives

$$f_k(u) \le k^n\int_{C_k(u)} f_k,\ \forall u \in A_k.$$

Thus $\mathcal{A}_k(f_k) \le 0$.

The extension constructed in Step 1 need not be convex outside P. We must therefore separate the points for which all convexity inequalities below are applied entirely inside P. Put

$$A_k^{\mathrm{deep}} := \left\{u \in A_k \mid u + [-k^{-1}, k^{-1}]^n \subset P\right\}, \qquad E_k := A_k \setminus A_k^{\mathrm{deep}}.$$

Every point of E_k has distance at most $1/k$ from ∂P. Moreover, the cell centered at such a point is contained in the boundary strip of width $3/(2k)$. These cells are disjoint and the strip has volume $\mathrm{O}(k^{-1})$. Hence

$$|E_k| \le C_P k^{n-1}. \tag{A.7}$$

For $u \in A_k$, set

$$D_{k,u} := k^n \int_{C_k(u)} f_k(x)\,\mathrm{d}x - f_k(u).$$

Jensen's inequality (which is valid here because $C_k(u) \subset P$) and the common continuity modulus give

$$0 \le D_{k,u} \le k^n \int_{C_k(u)} |f_k(x) - f_k(u)|\,\mathrm{d}x \le \hat{\omega}\left(\frac{1}{2k}\right). \tag{A.8}$$

It follows from (A.7) that

$$0 \le \frac{1}{k^{n-1}} \sum_{u \in E_k} D_{k,u} \le C_P \hat{\omega}\left(\frac{1}{2k}\right) \longrightarrow 0. \tag{A.9}$$

It remains to consider $u \in A_k^{\mathrm{deep}}$. For such u, define

$$h_u(x) := f_k(u+x) + f_k(u-x) - 2f_k(u)$$

whenever $u+x, u-x \in P$. On this symmetric domain, convexity of f_k implies that h_u is convex and $h_u(x) \ge h_u(0) = 0$. Since $u + [-k^{-1}, k^{-1}]^n \subset P$, all the following evaluations occur inside P. Symmetry of $C_k(0)$ and convexity give

$$\begin{aligned}
D_{k,u} &= \frac{k^n}{2} \int_{C_k(0)} h_u(x)\,\mathrm{d}x \\
&\le \frac{1}{2} \max_{v \in \{\pm(2k)^{-1}\}^n} h_u(v) \le \frac{1}{2} \sum_{v \in \{\pm(2k)^{-1}\}^n} h_u(v) \\
&\le \frac{1}{4} \sum_{v \in \{\pm(2k)^{-1}\}^n} h_u(2v) = \frac{1}{4} \sum_{w \in \{\pm k^{-1}\}^n} h_u(w).
\end{aligned} \tag{A.10}$$

In the third inequality we used $2h_u(v) \le h_u(0) + h_u(2v) = h_u(2v)$.

Fix $w = d/k$, where $d \in \{\pm 1\}^n$. The vector d is primitive, so the lattice lines parallel to d are indexed by the quotient $\mathbb{Z}^n/\mathbb{Z}d$. The set

$$P_k^{\mathrm{deep}} := \left\{x \in P \mid x + [-k^{-1}, k^{-1}]^n \subset P\right\}$$

is convex, and $A_k^{\mathrm{deep}} = P_k^{\mathrm{deep}} \cap k^{-1}\mathbb{Z}^n$. Consequently, the intersection of A_k^{deep} with each such lattice line is either empty or a finite consecutive chain

$$u_0, u_1, \ldots, u_m, \qquad u_{j+1} = u_j + w.$$

Choose a surjective integral linear map $\pi : \mathbb{Z}^n \to \mathbb{Z}^{n-1}$ with kernel $\mathbb{Z}d$, and extend it linearly to $\mathbb{R}^n$. Each chain determines a distinct point $\pi(ku) \in \pi(kP) \cap \mathbb{Z}^{n-1}$. Since $\pi(P)$ is bounded, this last set contains $\mathrm{O}(k^{n-1})$ lattice points. Hence the number of nonempty chains is at most $C_P k^{n-1}$.

Define

$$\delta_w f_k(z) := f_k(z) - f_k(z-w).$$

For a chain as above, the second differences telescope:

$$\begin{aligned}
\sum_{j=0}^{m} h_{u_j}(w) &= \sum_{j=0}^{m} \left(\delta_w f_k(u_j + w) - \delta_w f_k(u_j)\right) \\
&= \delta_w f_k(u_m + w) - \delta_w f_k(u_0).
\end{aligned}$$

The four points occurring in the last line belong to P, by the definition of A_k^{deep}. Therefore the common continuity modulus gives

$$0 \leq \sum_{j=0}^{m} h_{u_j}(w) \leq 2\hat{\omega}\left(\frac{1}{k}\right).$$

Summing over the at most $C_P k^{n-1}$ nonempty chains yields

$$0 \leq \sum_{u \in A_k^{\mathrm{deep}}} h_u(w) \leq C_P k^{n-1} \hat{\omega}\left(\frac{1}{k}\right). \tag{A.11}$$

Combining (A.10) and (A.11), and summing over the 2^n possible directions w, gives

$$0 \leq \frac{1}{k^{n-1}} \sum_{u \in A_k^{\mathrm{deep}}} D_{k,u} \leq C_P \hat{\omega}\left(\frac{1}{k}\right) \longrightarrow 0.$$

Together with (A.9), this proves

$$0 \leq -\mathcal{A}_k(f_k) = \frac{1}{k^{n-1}} \sum_{u \in A_k} D_{k,u} \longrightarrow 0,$$

and completes the proof. □

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